

Finger-Vein Biometric Recognition: A Review

Borui Hou^{ID}, Huijie Zhang^{ID}, and Ruqiang Yan^{ID}, *Fellow, IEEE*

Abstract—Finger-vein-based biometric technology has received wide attention and has made some positive achievements in improving the performance of personal identification. This article intends to provide an overview on the existing finger-vein-based biometric methods from the four steps of finger-vein recognition methods, including image acquisition, image preprocessing, feature extraction, and matching. After brief introduction of image acquisition and preprocessing, the applications of feature extraction method are summarized as: template-based method, representation-based method, and learning-based method. Moreover, deep-learning-based methods have been analyzed and discussed. To review and summarize these methods, the case studies of the finger-vein database are introduced to verify their effectiveness. Meanwhile, this article discussed the challenges and prospects of finger-vein recognition research, especially the 3-D finger-vein recognition research, to guide the future research direction in this field.

Index Terms—Biometrics, deep learning, feature extraction, finger vein, image acquisition, matching.

I. INTRODUCTION

BIOMETRICS, widely used for security authentication, is defined as the automated identity authentication through human biological or behavioral characteristics. Generally, behavioral characteristics include gait, voice, signature, and so on, while biological characteristics can mainly be divided into two categories, external biological characteristics (including face [1], [2], palm shape, fingerprint, and iris [3]) and internal biological characteristics (including palm vein, finger-vein, and dorsal vein). The widely used behavioral-based and external biological characteristics-based biometric methods are easily forged, while iris and DNA are inconvenient to be detected. As one of the most promising biometric technologies, the finger-vein-based biometric method has attracted widespread attention because of its high security, advantages on living identification, and high specificity.

Finger-vein-based biometric recognition mainly depends on the vein pattern image, which can be used to obtain the finger-vein features. A basic finger-vein capture device includes near-infrared (NIR) lights (700–1000 nm) and

a special charge-coupled device (CCD) camera. Since deoxyhemoglobin in the blood absorbs NIR light, a series of dark lines, i.e., vein patterns, appear in the finger image irradiated by NIR light. The captured image is then stored as a template of personal biometric authentication data after image preprocessing and feature extraction. During authentication, the finger-vein feature is extracted and compared with the stored user template. The finger-vein biometric recognition has several advantages as follows.

- 1) *Unique*: It is proved that the diversity of finger vein is similar to the iris pattern, which means finger-vein recognition is with high security and suitable for biometric authentication [4].
- 2) *Living*: The conditions of capturing vein patterns determine that it is only effective in the living body [5]. It is impossible to steal dead people's vein patterns.
- 3) *Permanent*: The vein pattern will not change in adulthood, which means it is permanent [6].
- 4) *Unforgeable*: The finger-vein pattern exists inside of the skin layer; therefore, it is hard to be damaged and replicated [7].
- 5) *Contactless*: Usually, the finger-vein images are captured by contactless sensors, which ensure hygiene and convenience [6].
- 6) *Robustness*: Kauba *et al.* [8] and Kauba and Uhl [9] investigated the impacts of different distortion types to finger-vein recognition performance. It turns out that model motion blur or defocuses, sensor aging, and related pixel defects do not affect too much. The compression impact of lossy has also been discussed in [10], because it is common to store template finger-vein image in recognition. It is found that the impact of applying compression standards is highly dependent on the used method, and JPEG2000 and JPEG XR perform well in different types of template-based methods.

Regardless of these advantages discussed above, the finger-vein-based biometric method is far from being perfect. Various internal and external factors may affect performance [11]. The configuration of the capture device is one of the most important internal facts that affect the recognition performance. External factors consist of ambient light, dust, humidity, electromagnetic radiation, temperature, illumination, humidity, and misplacement of the finger. Researches show that most of the internal and external factors can be ruled out by tuning device settings or applying processing strategies. For example, the impacts of ambient light can be reduced by adding an NIR pass-through filter, and the displacement of the finger can be solved by providing a finger position guidance strategy. In fact, lots of finger-vein recognition methods have been proposed to overcome those issues and improve recognition performance.

Manuscript received 30 March 2022; revised 20 July 2022; accepted 7 August 2022. Date of publication 18 August 2022; date of current version 8 September 2022. This work was supported by the IEEE Instrumentation and Measurement Society Graduate Fellowship Award. The Associate Editor coordinating the review process was Hongrui Wang. (*Corresponding author: Ruqiang Yan.*)

Borui Hou is with the Key Laboratory of Intelligent Computing Technology, Shanghai Aerospace Electronics Technology Institute, Shanghai 201109, China (e-mail: rex92hbr@163.com.cn).

Huijie Zhang is with the School of Biological Science and Medical Engineering, Southeast University, Nanjing 210096, China (e-mail: 230198906@seu.edu.cn).

Ruqiang Yan is with the School of Mechanical Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: yanruqiang@xjtu.edu.cn).

Digital Object Identifier 10.1109/TIM.2022.3200087

1557-9662 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

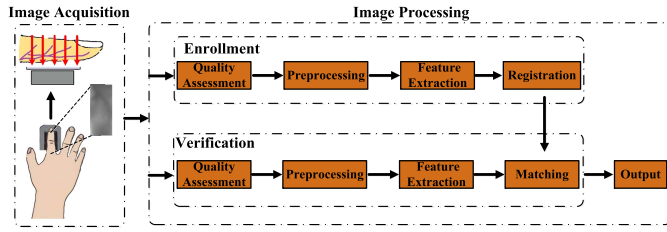


Fig. 1. Basic framework of finger-vein-based recognition system.

Since the finger-vein-based biometric system was proposed, many significant achievements have been made in this area as new systems have been continuously designed and deployed over the years. The basic framework of finger-vein recognition systems is shown in Fig. 1. Syazana-Itqan *et al.* [6] and Al-Tamimi [12] provided a detailed survey on finger-vein recognition in 2016 and 2019, respectively, and they focused on the general introduction and gave several future research directions for researchers to start the studies of finger-vein recognition. Shaheed *et al.* [13] also systematically summarized techniques in finger-vein recognition and spoof attacks in 2018. Besides, Dev and Khanam [14] presented a systematic introduction of the research works about finger-vein feature extraction methods. However, we observe that more and more literatures begin to pay close attention to finger-vein recognition in recent years, but without providing systematic guidance. Therefore, this article attempts to overview the latest algorithms and systems on finger-vein recognition and guide researchers in this field to carry out relevant studies. In this article, some key applications in finger-vein recognition will be discussed, including the following.

- 1) Summary and analysis of currently designed finger-vein measurement devices, including those for both research and commercial applications. In this article, we compared the existing finger-vein measurement devices in detail and divided them into four categories based on illumination type, top-light illumination, side-light illumination, bottom-light illumination, and both-light (side and top) illumination. Moreover, the finger-vein measurement device designed by ourselves is illustrated.
- 2) Overview of finger-vein feature extraction algorithms, especially deep-learning-based algorithms. This article gives a comprehensive summary of the existing feature extraction methods. In particular, the handcrafted feature extraction methods are divided into three categories: template-based method, representation-based method, and learning feature-based method. Meanwhile, recent proposed deep-learning-based finger-vein recognition methods are also revisited. Moreover, this article compared and analyzed the advantages and disadvantages of these existing methods through performance analysis on the public databases.
- 3) Trends and challenges of finger-vein biometric recognition. This article discussed the challenges and prospects of finger-vein recognition research, especially the 3-D finger-vein recognition research, to guide the future research direction in this field.

The remainder of this article is organized as follows. The introduction of publicly finger-vein databases is provided in Section II. Then, the existing handcrafted feature-based methods are analyzed in Section III, followed by the introduction of the deep-learning-based method in Section IV and the comparison discussion and prospects for finger-vein recognition in Section V. The conclusions are finally drawn in Section VI.

II. PUBLIC DATABASE

There are several publicly available finger-vein databases, including Hong Kong Polytechnic University (HKPU) database [15], Shandong University (SDU) database [16], University Sains Malaysia (USM) database [17], MNCBNU database [18], University of Twente Finger Vein Program (UTFVP) database [19], VERA database [20], [21], Tsinghua University Finger Vein and Finger Dorsal Texture Database (THU-FVFD) database [22], PLUSVein-FV3 Database [23], and GERWIN Database. The summary of those publicly available databases is listed in Table I. The examples of these introduced finger-vein databases are as shown in Fig. 2. Except for the VERA database, each of these databases provides more than 1000 samples for finger-vein recognition. The VERA database is usually analyzed in finger-vein spoofing. Some of the databases are captured in two sessions, which can be used to discuss the robustness of finger-vein recognition methods. Moreover, not all samples in these databases are of high quality. In general, the images from the HKPU database, USM database, and UTFVP database are acquired in high quality, and the vein pattern networks are clearly shown in these images. The details of these public finger-vein databases are listed below.

- 1) *HKPU Database [15]*: The HKPU database contains 3132 samples from 156 volunteers and is provided by HKPU. In this database, samples from 105 volunteers are collected in two sessions. In one session, each volunteer has the images of the index and middle finger of the left hand. The collection for each finger is repeated six times. Other 51 volunteers provide samples in one session. The original samples are stored in gray-level bitmap (BMP) format with a resolution of 513×256 pixels.
- 2) *SDU Database [16]*: The SDU database contains 3816 samples from 106 volunteers and is provided by SDU. In this database, samples from 106 volunteers are captured in one session, and each volunteer has the images of the index, middle, and ring fingers of both hands. The collection for each finger is repeated six times. The original samples are stored in gray-level BMP format with a resolution of 320×240 pixels.
- 3) *USM Database [17]*: The USM database contains 5904 samples from 123 volunteers and is provided by USM. Samples are captured in two sessions with six times per finger in each session. The original samples are stored in gray-level BMP format with a resolution of 640×480 pixels.
- 4) *MNCBNU Database [18]*: The MNCBNU database contains 6000 samples from 100 volunteers who are from 20 countries with different races. In this database,

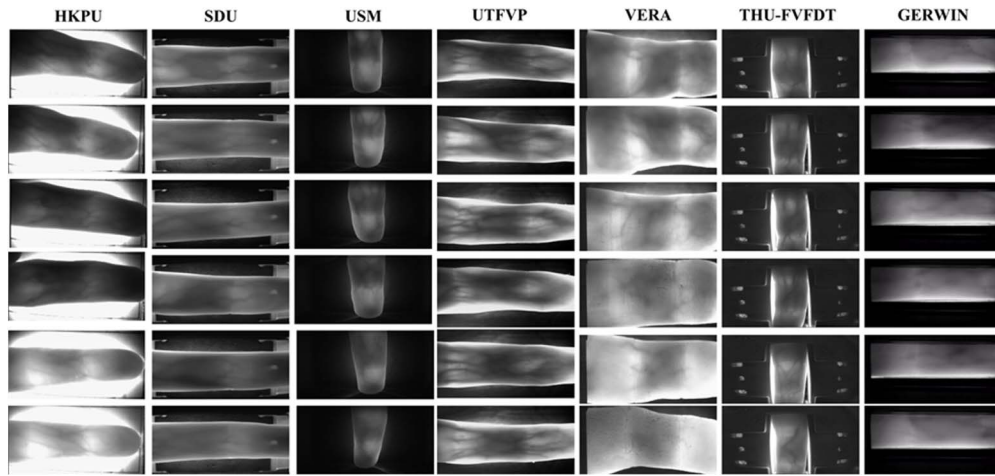


Fig. 2. Demo images from public finger-vein image databases.

TABLE I
SUMMARY OF FINGER-VEIN DATABASE

Ref	Database	Finger No.	No. of Image per Finger	Session	Size	Total Image	Illumination type
[15]	HKPU	312	6/12	1/2	513×256	3132	top-light illumination
[16]	SDU	636	6	1	320×240	3816	top-light illumination
[17]	USM	492	12	2	640×480	5904	top-light illumination
[18]	MMCBNU	600	10	1	640×480	6000	top-light illumination
[19]	UTFVP	360	2	2	672×380	1440	top-light illumination
[20,21]	VERA	220	2	1	665×250	440	top-light illumination
[22]	THU-FVFD1	220	2	1	720×576	440	top-light illumination
	THU-FVFD2	610	2	2	720×576	1220	
	THU-FVFD3	610	8	2	720×576	4880	
[23]	PLUSVein-FV3	360	20	1	1280×1024	7200	/
/	GERWIN	1386	6	1	600×200	8316	side-light illumination

the samples from 100 volunteers are captured in one session, and each volunteer has the images of index, middle, and ring fingers of both hands. The collection for each finger is repeated ten times. The original images are stored in gray-level BMP format with a resolution of 480×640 pixels.

- 5) *UTFVP Database [19]*: The UTFVP database contains 1440 samples from 60 volunteers and is provided by the University of Twente, The Netherlands. Samples are acquired in two different sessions with two images per finger in every session, and each volunteer has the images of index, middle, and ring fingers of both hands. The original samples are stored in gray-level PNG format with a resolution of 672×380 pixels.
- 6) *VERA Database [20], [21]*: The VERA database contains 440 samples from 110 volunteers and is provided by Idiap Research Institute and Haute Ecole Spécialisée de Suisse Occidentale in Sion, Switzerland. There are 70 males and 40 females in this database. Samples are captured in two sessions with two times per finger in

each session, and each volunteer has the images of index finger of both hands. The original images are stored in gray-level BMP format with a resolution of 665×250 pixels.

- 7) *THU-FVFD Database [22]*: The THU-FVFD database is provided by Tsinghua University and has three subdatabases.

- a) THU-FVFD1 contains 440 samples from 220 volunteers. Images are captured in two different sessions with intervals of about dozens of seconds, and each session has two images. The original samples are stored in gray-level BMP format with a resolution of 720×576 pixels.
- b) THU-FVFD2: It is an extension of THU-FVFD1 database. The first 440 samples are from the THU-FVFD1 database, and the remaining samples are collected from 390 volunteers in two sessions. The collected images are stored as regions of interest (ROI) in gray-level BMP format with a resolution of 200×100 pixels.

- c) THU-FVFDT3 database is the expanding THU-FVFDT2 database. It contains 4880 samples, which are captured in two sessions with four images per finger.
- 8) *PLUSVein-FV3 Database* [23]: The PLUSVein-FV3 database contains 60 volunteers and is provided by the Computer Sciences Department of the University of Salzburg. It consists of four subsets: one dorsal and one palmar finger-vein subset captured using transillumination with the LED and the laser module-based scanner, respectively. Samples are acquired five times per finger in one session, and each volunteer has the images of index, middle, and ring fingers of both hands. Each subset contains 1800 samples and a total of 7200 samples for the whole database. The original samples are stored in gray-level PNG format with a resolution of 1280×1024 pixels. The extracted ROI images are stored in a resolution of 736×192 pixels.
- 9) *GERWIN Database*: The GERWIN database contains 8316 samples from 231 volunteers and is provided by our designed finger-vein capture device. The main purpose of collecting this database is to provide a finger-vein database with more samples for researchers. In this database, all the volunteers are under 21 years old, and the samples are captured in one session with six times per finger. Each volunteer has the samples of index, middle, and ring fingers of both hands. The collected images are stored as ROI images in gray-level JPG format with a resolution of normalized to 600×200 pixels.

III. STEPS OF FINGER-VEIN RECOGNITION

The typical finger-vein biometric recognition technique includes image acquisition, image preprocessing, feature extraction, and matching.

A. Image Acquisition Technology

As the basic step, image acquisition technology is directly related to image quality. The vein patterns are subcutaneous and hidden, which require special acquisition technology to make them visible. Generally, four methods can be used to acquire the vein pattern in medical imaging: X-ray, ultrasonic [24], lasers [23], [25], [26], and infrared [27], [28], [29], [30], [31]. The X-ray and ultrasonic techniques are usually applied in the medical field, and ultrasonic has been introduced in palm vein recognition but not for finger-vein recognition yet. Lee *et al.* [25] claimed that the laser-based method improves performance by 60%, compared with the infrared-based method. Showing the best contrast in finger-vein and blood flow image, the laser-based method is usually proposed for liveness detection and anti-spoofing authentication [26]. In finger-vein recognition, most of the researchers choose the infrared method to acquire vein patterns [31]. There are two choices of infrared image, the far-infrared (FIR) image and NIR image. Compared with FIR images, the NIR images contain more useful information. Therefore, NIR light with a wavelength of 700–1000 nm is widely used for image acquisition. The impact of NIR light wavelength has been discussed

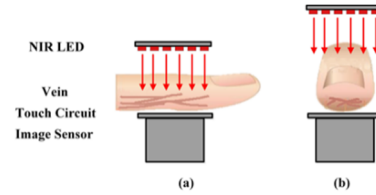


Fig. 3. Top-light illumination type. (a) Side view. (b) Front view.

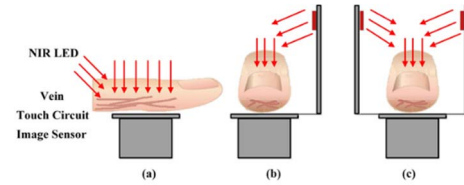


Fig. 4. Side-light illumination type. (a) Side view. (b) Front view of one side. (c) Front view of both sides.

in [32], and research shows that the best performance is obtained in the wavelength range of 875–890 nm.

Based on the illumination type, the NIR-based finger-vein image acquisition can be divided into four categories: top-light illumination [15], side-light illumination [19], bottom-light illumination [30], and both-light (side and top) illumination [33]. The top-light illumination method (as shown in Fig. 3) takes advantage of light transmission to capture the image with high contrast because the light source, usually LED, locates upon the finger to make the light pass through the finger and has few impacts on ambient light reflections. The disadvantage is that the users need to put their fingers into the device, which causes discomfort and inconvenience. The light source of the side-illumination method (as shown in Fig. 4) is on the side of the finger (one side or both sides), which makes use of light reflection. This design makes the capture device more compact and the users can see their fingers during the operation. The disadvantage is that the side of the fingers is usually overexposed [19]. The bottom-light illumination method (as shown in Fig. 5) proposed in [30] uses a circle of multiple NIR LEDs around the camera to capture finger-vein images, thus makes it become one of the smallest devices and can be used for the smartphone. The disadvantage is that this method cannot provide good quality images as compared to other methods.

The light source of the both-light illumination method is on the side and above the finger, which makes use of both light reflection and light transmission. Experience in [33] explored the effectiveness of different illumination types (top, side, and both), and results indicated that the both-light illumination method achieves the best performance. This method combines the advantages and disadvantages of top-light illumination and side-light illumination methods.

The works on the design of image capture devices are carried out by the industrial stakeholders for commercial application and by academic researchers to improve the finger-vein image quality and recognition performance. The commercial capture devices have been used in a wide range of application

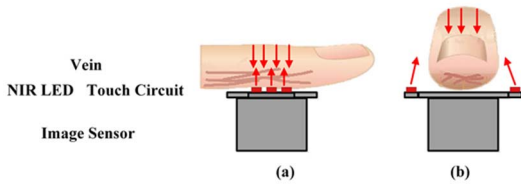


Fig. 5. Bottom-light illumination type. (a) Side view. (b) Front view.

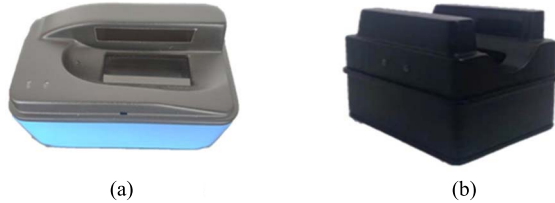


Fig. 6. Pictures of our designed finger-vein image capture device. (a) Academic use. (b) Commercial use.

scenarios, from securing a personal computer over additional authentication to high-security access control systems. Depending on the compact and user-friendly design, the finger can be easily located and then preprocessed for recognition. However, most commercial finger-vein recognition devices have a major drawback: the generated templates are stored on the local device and cannot be used in complex scenarios. The templates registered on a single device cannot be used on other devices. Once the device is replaced, the template needs to be re-registered. The major companies providing finger-vein recognition devices are Hitachi, Mofiria, IDEMIA, M2SYS, Athena Eyes, MYID, SDKJ, and GERWIN. They provide automatic finger position adjustment for comfortable usability and support two types of authentication processes, both in the device and on the server. The commercial finger-vein devices are suitable for semi-outdoor application scenarios and can be used in security, education, public security, human resources, finance, medical, and other industry solutions. Using the technology provided by Hitachi, Sthaler Ltd. developed a biometric payments systems called Fingopay. The Fingopay system allows customers to use their finger-vein patterns for payment. For large-scale finger-vein recognition scenarios, GERWIN Ltd. developed a finger-vein recognition system, including an image capture device and corresponding server. The data obtained by different devices are compared with template in the server, which improves efficiency and safety. As these are commercial products, details about their design are limited except for the recognition performance and appearance information. The important information of commercial finger-vein devices is the performance that includes false acceptance rate (FAR), false reject rate (FRR), and failure to enroll (FTE). Furthermore, all commercial devices are compact, fast transaction times, and working with corresponding software. The pictures of our designed finger-vein capture devices are shown in Fig. 6, and the characteristics are listed in Table II.

The academic versions of capture devices have a variety of different designs and characteristics for research. The characteristics of the proposed academic finger-vein capture devices

TABLE II
CHARACTERISTICS OF COMMERCIAL FINGER-VEIN DEVICES

Company	Illumination Type	Performance
Hitachi	top-light illumination	FAR: 0.001% FRR: 0.01% FTE: 0.03%
Mofira	side-light illumination	/
IDEMIA	side-light illumination	FAR:0.01% FRR:0.001%
M2SYS	top-light illumination	FAR:0.0001% FRR:0.01% FTE:0%
Athena Eyes	side-light illumination	/
MYID	side-light illumination	/
SDKJ	side-light illumination	FAR:0.00001% FRR:0.01%
GERWIN	side-light illumination	FAR:0.0034% FRR:0.17%

TABLE III
CHARACTERISTICS OF ACADEMIC FINGER-VEIN DEVICES

Ref	Illumination Type	Sensors	Light wavelength
[15]	top-light illumination	NIR camera	850mm
[17]	top-light illumination	Sony PSEye camera	850mm
[18]	top-light illumination	/	850mm
[19]	top-light illumination	monochrome CMOS sensor	850mm
[23]	top-light illumination	monochrome CMOS sensor	808mm
[25]	top-light illumination	IR image sensor	830mm
[26]	top-light illumination	IR image sensor	830mm
[27]	top-light illumination	Infrared web camera	850mm
[28]	top-light illumination	NIR camera	808mm
[29]	top-light illumination	NIR camera	850mm
[30]	bottom-light illumination	Monochrome Camera	940mm
[33]	both-light illumination	NIR camera	920mm
[34]	top-light illumination	grayscale 1/3-inch CMOS camera	/
Our device	Side-light illumination	CMOS camera	850 nm

are listed in Table III. The majority of academic devices are larger than commercial devices, except those for miniaturized device research [30]. The interesting characteristics of all these devices are as follows.

- 1) All capture devices have used sensors with a filter (NIR pass) and a lens. The quality of sensors is varied from expensive industrial cameras [19] to low-cost Web cameras [15].
- 2) The top-light illumination method is the most popular method to make public databases for academic studies. However, the side side-light illumination method, which has a compact design, is widely used in commercial scenarios, because it is more user-friendly and it is convenient for users to adjust the placement of the finger.
- 3) Some devices have a large image capture area, and it is necessary to determine the finger area to reduce the impact of finger movement. The determination of the finger area affects the reliability and accuracy of the recognition. Although reasonable efforts have been made in the development, capture devices still have some problems, such as low contrast, noise, rotation, and movement changes. These problems rely on image preprocessing methods to solve.

B. Image Preprocessing

After image acquisition, the image preprocessing techniques are applied to improve the image quality. Many factors have impacts on finger-vein image acquisition, such as ambient light, dust, or dirt on the sensor, humidity, temperature, illumination, humidity, and user behavior. The captured image will be of poor quality if these factors are not well-taken care and will lead to the poor performance of the recognition. Therefore, several image preprocessing techniques have been designed to relieve these problems. The image processing methods include image quality assessment, image restoration, ROI extraction, image enhancement, and normalization.

1) *Image Quality Assessment*: The image quality assessment has many important roles in finger-vein-based biometric systems. During the acquisition operation, the measurement of image quality can help monitor and control the quality of the captured images [35]. The results of finger-vein image quality also help choose the proper image processing method and adjust the steps of the recognition method to improve the performance. Based on human visual system sensitivity and prior knowledge, many handcrafted descriptor-based methods were introduced in image quality assessment. For example, Ma *et al.* [36] proposed a signal-to-noise-ratio (SNR)-based human visual system [human visual signal-to-noise-ratio (HSNR)] for image quality measurement. The specially designed HSNR combines effective area score, finger shifting score, and contrast score and can be used as the guidance of finger-vein image acquisition. Using the three most effective quality scores from five human visual system-based image quality assessment methods, a multi-image quality assessment (MQA) method was proposed in [37], and good performance was achieved in the USM database with an equal error rate (EER) of 0.130%. The limitation of the score fusion-based quality score method is that the quality scores are all evaluated in a linear model, which cannot comprehensively evaluate the image quality [38].

Machine learning-based methods were also used to assist in the determination of handcrafted descriptor-based image

quality evaluation. Zhou *et al.* [38] estimated the image quality by introducing the support vector regression (SVR) to establish the relationship between quality scores and compound features including local features and global features. Yang *et al.* [39] utilized a support vector machine (SVM)-based image quality evaluation method, where three different quality features were selected and then classified by SVM to measure the image quality. Considering the specificity of finger-vein recognition that most of the methods were applied to grayscale and binary images, Qin *et al.* [35] used hybrid features from the grayscale and binary image to represent the image quality, and then, SVM was proposed to identify the low-quality images. However, these handcrafted-descriptor-based methods have many issues. First, some features are directly based on the human visual system and prior knowledge, which cannot properly represent the finger-vein image quality. Second, single features cannot comprehensively evaluate the finger-vein image quality, while the multifeature methods are time-consuming and complex. Moreover, it is hard to choose useful features and evaluate them in a proper model.

Based on the methods of human visual system sensitivity and prior knowledge, image quality assessment methods usually use the statistical probability model between image features and image quality to obtain the posterior probability of the image as an evaluation score. The choice of appropriate probabilistic mathematical model and sample size is the key to the evaluation performance. Without prior knowledge, machine learning methods can learn to establish a mapping relationship between image features and image quality, thereby judging the quality of new images. However, machine learning methods rely on large amounts of training data and precise data labels.

2) *Image Restoration*: In finger-vein imaging, the NIR light passing through human fingers can be absorbed, refracted, and scattered by biological tissues. Regarded as a complex heterogeneous optical medium, biological tissue affects the light transmission, while the lights suffer from scattering. Similar to the light propagating in the fog, the NIR light scattering in biological tissue has significant impacts on the clarity resolution of the image. Therefore, the research of image restoration in finger-vein recognition mainly focuses on scattering removal. Moreover, the broken vein pattern image restoration, which is caused by contamination in image acquisition, is also discussed in the study.

Based on prior knowledge of biomedical imaging, there are lots of image restoration methods for scattering removal, such as point spread function (PSF) [7], [40], [41], [42], biological optical model (BOM) [42], [43], [44], dehazing-based technique [45], [46], [47], bilayer diffusion model [48], and direction-variance-boundary constraint search (DVBCS) model [49]. The PSF model was first introduced in finger-vein image restoration in [7], and then, many variants of the PSF model, such as depth-PSF [41], two depth-PSF [40], Gaussian-PSF [40], and multilayered PSF [42], have been proposed for image restoration. Experiments show that the PSF-based model can greatly suppress the effects of image blurring but not measure the blurring effect of microscopical propagation [42]. Yang *et al.* [43] estimated and properly

removed the light scattering by applying the BOM model. The specially designed BOM model improved the image contrast and thus achieved high performance in matching. The limitation of the BOM model is that it is only designed for homogenous and not for heterogeneous medium. The hybrid method which takes advantage of both the PSF model and BOM model was proposed in [42], and the multilayer PSF and BOM were integrated to remove light scattering and achieved good performance in image restoration. Similar to the light scattering in fog, dehazing technology can be introduced in the scattering removal of image restoration. Yang *et al.* [45] and Yang and Shi [46], [47] applied Koschmieder's law-based dehazing technology to remove the scattering and make the vein image clearer. However, the contrast between the vein region and background is still low after image restoration. You *et al.* [48] described the light scattering in finger skin tissue through a bilayer diffusion model by taking the whole finger tissue into consideration, and then, a multi-Wiener steins unbiased risk estimate and linear expansion threshold (SURE-LET) method was used for image restoration. The proposed method is a kind of light scattering removal method and comprehensively considers the whole finger issue. The DVBCS model was established in [49] for restoration, where the broken branches of finger-vein patterns were restored effectively and rich vein pattern features were provided for matching. These studies on finer vein image restoration mainly focused on blur restoration problems, and experiments were conducted on databases collected by a specific image acquisition device. The predetermined constant parameters of these models may not be suitable for the different devices.

These studies on finger-vein image restoration mainly focused on blur restoration problems, and experiments were conducted on databases collected by a specific image acquisition device. The predetermined constant parameters of these models may not be suitable for different devices. To deal with the light scattering issue in biological tissues, many biomedical imaging models are introduced into finger-vein image restoration. These methods consume a lot of computing resources but fail to significantly improve the performance of finger-vein image recognition.

3) *ROI Extraction*: The captured raw image includes two parts: the valuable vein pattern-related region (denoted as foreground) and the black region (denoted as background). The process of ROI extraction is to locate the finger-related region from the raw image and delete useless information. Some image alignment methods are also applied during the ROI extraction to reduce the effects of finger movement and rotation. The details of the ROI extraction method are introduced step and by step and summarized as follows.

1) *Boundary Selection*: Several image segmentation methods, such as fixed window-based method [45], [46], [50], [51], edge operator-based method [52], [53], [54], mask-based method [56], [57], [58], [59], threshold-based method [15], [60], and superpixel-based method [61], [62], were deployed to distinguish the finger-related region from the background. The fixed window [45], [46], [50], [51] was usually adopted to extract the candidate region when the position of the finger in the image

is generally fixed. Yang *et al.* [51] proposed the fixed window to select a candidate finger-vein region and then adjust the two boundaries of the finger by computing the standard deviation map (SDM). The shortcoming of this simple and practical method is that it cannot adapt to the changes in image size and is only suitable for images collected by a specific device. Some edge operator-based methods [52], [53], [54] were introduced to detect the finger boundaries since the distribution of background and finger area is different. Sobel operator, which is sensitive to noises, was used to distinguish the shape of the finger.

The predefined mask was proposed in [56], [57], [58], and [59] to detect the finger boundary. The utilized mask was filtered with the first or second half of the original image column by column to select the pixel with the maximum value in one column, and then, the finger boundary was detected by these maximum pixels. However, the edge operator-based and mask-based methods performed badly in the image with lots of noises. The pixel values of finger area and background are different, so that the threshold-based method [15], [60] was applied for finger-vein candidate region selection. Liu *et al.* [60] used the classical Otsu threshold to segment the finger area. In fact, it is hard to choose a proper threshold, because the gray value of the local finger area is close to that of the background. A superpixel-based method was presented in [57] to detect the finger boundary by using the simple linear iterative clustering (SLIC) method. Compared with other methods discussed above, the superpixel-based method can be used for images collected by multiple devices and is robust to image variations, such as image size and background noise.

2) *Image Alignment*: To resist the finger rotation interference, image alignment methods, such as centerline-based method [54], [57], [58], [60], [61], [63], pattern normalization [34], affine transform-based method [56], [58], geometric analysis-based detection and correction (GADC) method [62], Hough transform-based method [64], and iterative closest point (ICP) method [65], are presented to adjust the finger images. After determining the finger boundaries, the centerline of the finger was calculated by the least-squares method in [54], [57], [58], [60], [61], and [63], and then, the angle between the centerline and standard direction was used for image adjustment. Huang *et al.* [34] used the pattern normalization method to remedy the misalignment of the finger based on the assumption that the cross section of the finger is roughly in the shape of an ellipse and the captured image is close to the surface. Lee *et al.* [56] adopted an alignment method based on the minutia point of finger-vein image and the alignment model. Experiments show that the original image can be successfully aligned by the proposed affine model. Chen *et al.* [62] classified the finger deformation into three types: finger tilt, finger bending, and longitudinal finger rotation. Then, the linear transformation

and nonlinear transformation were introduced to correct the image. Xie *et al.* [64] utilized a Hough transform-based method for rotation correction, which considered finger curvature itself. The ICP method, based on finger contour of 2-D image, was applied in [65], and it is more advance than the centerline-based method.

- 3) *ROI Localization*: The image localization methods are proposed to decrease the impacts of finger movement during authentication. The phalangeal joints, related to the middle of the phalanx of a finger, can be used for finger-vein ROI localization according to previous researches [45], [46], [50], [51]. More light passes through the clearance region between two cartilages during the finger-vein image acquisition because the density of cartilage is much lower than that of bones. Thus, the finger joints-related region shows brighter than the rest. The width of clearance of a finger interphalangeal joint is 1.5–2 mm and can be regarded as a pixel width line in a finger-vein image. There are two phalangeal joints, the distal interphalangeal joint and the proximal interphalangeal joint, which can be used as the localized benchmark, and the distal interphalangeal joint shows brighter. The maximum value of the row is usually selected to determine the position of the distal interphalangeal joint [45], [46], [50], [51]. Notice that the row with the highest pixel value does not always represent the phalangeal joint, Yang *et al.* [52] detected the phalangeal joint by calculating the sum of gray values in the sliding window, and the region between the distal interphalangeal joint and the proximal interphalangeal joint was selected as ROI region. Qiu *et al.* [53] proposed a dual-sliding window method for phalangeal joints detection, and the application of two nearest neighbor sliding windows ensured that the effects of light interference can be eliminated. A operator-based finger-vein region localization method was proposed in [55], where a large receptive field gradient operator was adopted to locate the ROI region. Zhang *et al.* [66] introduced the gradient algorithm in the finger-vein candidate region to analyze the grayscale distribution for interphalangeal joint detection. The finger texture and finger-vein pattern have a corresponding relationship when they are captured simultaneously, and the finger crease shows the finger position. The drawback of this method is that it has to capture finger texture images and finger-vein images at the same time.

Since the location of ROI is selected and the rotation of the finger image is corrected, the ROI image can be segmented from original images for further processing. The processes of ROI extraction resist the influence of finger movement and rotation, delete the useless information of background, and thus are helpful in feature extraction and matching. The example of the ROI extraction process is shown in Fig. 7. To emphasize the processing effect, the image from the HKPU database is used, and three typical methods, including boundary selection [56], ROI localization [51], and image alignment [34], are applied for ROI extraction.

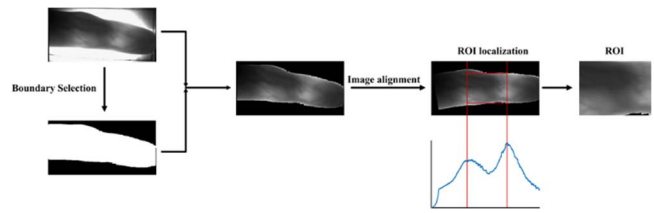


Fig. 7. Examples of ROI extraction, boundary selection method [56], ROI localization [51], and image alignment [34].

- 4) *Image Enhancement and Normalization*: Focused on contrast enhancement and denoising, the image enhancement methods are introduced to make the venous network clearer. The histogram equalization (HE)-based methods, such as HE [21], [28], [45], [67], adaptive HE (AHE) [19], [68], contrast-limited AHE (CLAHE) [53], [69], [70], [71], [72], parametric-oriented histogram enhancement (POHE) [37], [73], fuzzy HE (FHE) [74], [75], and modular AHE (MAHE) [63], have been adopted in the image enhancement. The HE, a simple and effective method, was presented in [67] for image contrast enhancement, while edge-preserving filter and elliptic high-pass filter were deployed for denoising. To optimize local image contrast, the AHE method was adopted in finger-vein image enhancement, and experiments showed that the AHE had enhanced a small region of the image, compared with basic HE [68]. The limitation of HE and AHE is that the noise is also enhanced during the processing. Therefore, the CLAHE method, which can effectively suppress noise and improve image contrast, was widely used in finger-vein image enhancement and achieved better performance. Hsia *et al.* [73] utilized POHE, which meets the requirement of local contrast enhancement and low computation time, in the finger-vein image to enhance image contrast. Jumaa and Zidan *et al.* [74] combined the fuzzy histogram and cumulative HE in finger-vein enhancement, while the cumulative HE was applied before FHE. Considering both image quality and time cost, the MAHE was proposed in [63] to improve the vein definition and suppress noise. In the finger-vein enhancement, the HE-based methods still need to be improved to meet the requirement of time consumption and image quality.

Some filtered-based methods, especially Gabor filter-based methods [44], [46], [59], were also introduced in image enhancement. For example, Yang and Shi [44], [46] adopted a multiscale multiplication rule (MSMR)-based method for enhancement. The designed MSMR comprehensively considered the vein information enhancement and noise suppression, and the results of MSMR significantly improved the appearance of vein networks. A fuzzy-based fusion method was proposed in [59], and the Gabor filters and Retinex filter were adopted to enhance the global and local characters of the image. Lee *et al.* [76] proposed a modified Gaussian high-pass filter, which has an advantage in extracting vein networks for image enhancement. The oriented filters were utilized in [77] to generate direction maps, and then, the extracted vein objects of these maps were used to enhance finger-vein images. Moreover, the pulse coupled neural network (PCNN) was introduced as an attractive alternative to previous methods for finger-vein image enhancement in [78], while a new strategy

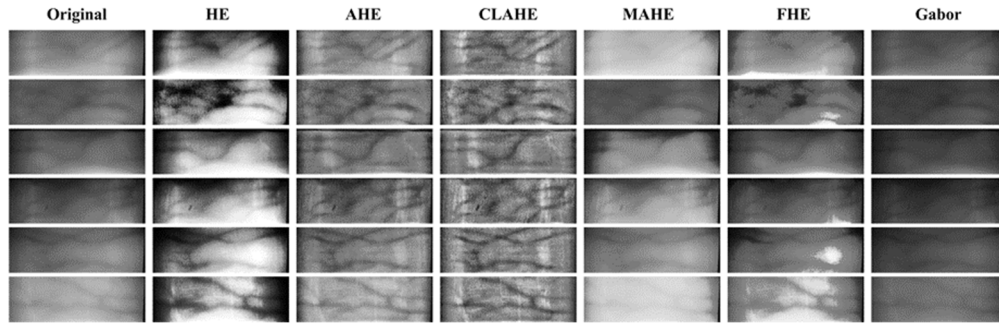


Fig. 8. Examples of image enhancement methods.

was designed for simplified PCNN model parameter settings without training. Experiments showed that the effectiveness of designed PCNN model and vein networks was enhanced. The examples of several image enhancement methods are shown in Fig. 8. The image from our collected GERWIN database is used, and six typical methods are proposed for comparison. The CLAHE method can better improve image contrast and highlight finger-vein patterns. The Gabor method performs well in noise suppressive, but not good at image contrast enhancement method.

The image normalization methods, such as illumination normalization [79], brightness normalization [80], and intensity variation (IV) normalization [81], were proposed to remedy the impacts of light intensity, personal finger thickness, and misalignment. Kim *et al.* [79] proposed an illumination normalization method to normalize the illumination. Vlachos and Dermatas [80] utilized a brightness normalization method that used a linear transformation scheme to correct the nonuniform illumination and shading artifacts, and then, a modified separable Mumford Shah model was used for enhancement. The IV normalization method was proposed in [81], while the proposed method focused on removing the impacts of IV and helped enhance the vein image.

C. Feature Extraction

After preprocessing, the feature extraction methods are adopted in the finger-vein image to generate unique and stable features for further recognition. The validity and reliability of the proposed feature extraction method can be directly related to the recognition performance. Generally, they can be divided into three categories: template-based method, representation-based method, and learning feature-based method.

1) *Template-Based Method*: In finger-vein feature extraction, vein pattern networks and minutia features, which represent the specificity of finger-vein images, are chosen to generate template images. The vein pattern-based method uses the vein networks as features to represent the finger image. Generally, there are three types of methods to extract finger-vein networks, i.e., valley-based method, local region-based method, and global image-based method.

The valley-based method assumes that the vein pattern points have a lower pixel value than other points on the cross-sectional profile, which looks like a valley. The valley-based methods, such as repeat line tracking (RLT) [5], [82], maximum curvature point [84], [85], and orientation map-guided

curvature [86], have been developed for vein pattern extraction. The RLT method was proposed in [5] to identify local dark vein lines, and Liu *et al.* [82] modified the RLT method by estimating the width of the finger vein first according to the binary and skeleton image. The maximum curvature point method [84] was utilized to detect the centerlines of veins by searching the curvature of the image profiles, and the Hessian enhancement was applied in [85] to remove noises during the process of the maximum curvature point method. Based on the analysis of vein anatomy structure [86], the orientation map-guided curvature was used to extract vein networks, and vein backbone was further extracted for vein network calibration to reduce the impacts of finger displacement.

The local region-based method, such as wide line detector (WLD) [34], matting-based segmentation method [47], difference curvature [87], and Gaussian weight spatial curve transformation (GWSCF) [88], usually distinguished the vein pattern related pointed by predefined local regions. In WLD [34], the point on the vein lines was extracted if the value of the pixel is higher than a fixed threshold in the local region. The matting-based segmentation approach was proposed in [47], while the K -means clustering method was used to obtain skeletons first, and then, the matting-based segmentation method preserved vein pattern shapes for finger-vein extraction. Qin *et al.* [87] introduced the difference curvature method in finger-vein extraction. Unlike the maximum curvature point method only focused on the curvature of the pixel, the difference curvature method makes use of the difference value of extracted curvatures and can effectively reduce the impacts of noise and irregular shading. Yang *et al.* [88] used a set of spatial curve filters (SCFs) to fit vein patterns closely and locally, and experiments demonstrated that the GWSCF method could effectively and adaptively extract vein patterns.

According to the distribution of the whole image, global-based methods have been proposed in vein pattern extraction. The mean curvature method utilized in [89] extracted the vein patterns by detecting valley-like structures with negative mean curvatures. Qin *et al.* [90] proposed Radon transform to obtain vein patches, where the vein patterns were enhanced and detected according to the curvature values in Radon space. Combined curvature and radon-like features, Yao *et al.* [91] enhanced vein patterns and suppressed noises to obtain smoothing and genuine vein structure for matching and verification. Zhang *et al.* [92] designed an active contour-based method for vein-pattern extraction, while the

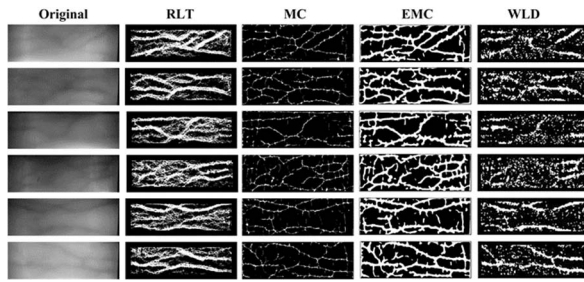


Fig. 9. Examples of vein pattern-based methods. RLT: repeat line tracking; MC: maximum curvature point; EMC: enhanced MC point; WLD: wide line detector.

dehazing algorithm was used for image enhancement and kernel fuzzy *C*-means (KFCM) algorithm was conducted for extraction procedure. Experiments proved that the proposed method can improve the verification performance.

The examples of several vein network extraction methods are shown in Fig. 9. The image from our collected GERWIN database is used, and four typical methods are proposed for comparison. It can be found that all these methods can successfully extract the hidden vein network information in the captured image, but the effect of these methods is different. The vein network extracted by the valley-based method is clear and contains less burr information. It is also necessary to point out that the width of the vein network obtained by different methods is different, and the image thinning method needs to be used for further processing. Generally, the vein pattern-based methods are unable to achieve promising results because of the low quality of the captured image and the limitations of algorithms. Most of the proposed methods can effectively remove the noises and useless regions, but few methods are robust to the variations of finger images. Furthermore, the generated templates usually contain rich features that have impacts on template storage and matching.

The minutia feature-based method detects the point patterns for matching. Different from the vein pattern-based method, the characteristic of the finger-vein is represented by some discrete minutia points, which can retain invariance and discrimination information and are robust against image variations. Yu *et al.* [93] extracted the minutia features, including bifurcation points and ending points, to represent vein pattern shape, and then, the modified Hausdorff distance was used for evaluation. Experiment results showed that these minutia features could robustly represent finger-vein images. The junction point-based curve tracing method was adopted in [89] for curve segment extraction, and then, curve segments were used to generate structure codes. The singular value decomposition (SVD)-based minutiae matching method (SVDMM) was proposed in [94], where the bifurcation point and ending point were extracted from the skeletonized binary finger-vein images, and then, coordinate value, local average intensity (LAI), local intensity deviation (LID), and local extensive binary pattern (LEBP) were analyzed for SVD-pairing matching. The specially designed SVDMM algorithm worked better in minutia pairs discovery and false pairs removing. Yang *et al.* [95] used the bifurcation point

and its local vein branches, named tri-branch vein structure, as minutia features for template matching. The shortcoming of the existing minutiae-based methods is that, when matching, the minutia features are sensitive to rotation, deformation, and translation, thus making false matching. Moreover, the minutia feature extraction of the existing minutiae-based method relies on skeletonized binary images, which is affected by the limitations of extraction methods.

2) *Representation-Based Method*: Different from the template-based method, the representation-based methods utilize descriptors, transformers, texture extraction, and dimensionality reduction for feature extraction. The features extracted by finger-vein image representation-based methods are then used for recognition.

Descriptor-based features have been widely used in feature extraction. For example, Khusnuliawati *et al.* [96] adopted a multifeature fusion method, where the scale-invariant feature transform (SIFT) and LBP were employed for feature extraction. The SIFT feature has resistance to rotation, deformation, and translation, while the LBP feature has an amount of local characteristics information. The local binary pattern (LBP) method [97] calculates the difference between the value of each center pixel and the value of its neighboring pixels in a gray image, which is represented by binary codes. Lee *et al.* [98] designed a weighted LBP method, where the extracted LBP features were divided into three categories and have different weights determined by recognition performance. Rosdi *et al.* [99] used a local line binary pattern (LLBP) to extract features. Different from LBP, the neighborhood of LLBP is a straight line. Experiments showed that the LLBP method worked better than LBP. As a variation of LBP, the binary gradient contour (BGC) method was used in [100]. Unlike LBP, the BGC compares the adjacent pixel value along the periphery of 3×3 neighborhoods and has better texture discrimination capability. The local hybrid BGC (LHBGC) method, as an improvement of the BGC method, was introduced in [101]. The sign and magnitude components of the finger-vein image were extracted by LHBGC to achieve high performance. Inspired by LBP, the personalized best bit map (PBBM) method was utilized in [102], and the LBP features from the same individual were calculated to generate best bit map codes for recognition. The personalized weight maps (PWMs) method [103] is another improvement of the original LBP method. Unlike the PBBM method which only used the “best bits” to represent the finger-vein image, the PWMs method introduced different weights for bits with different degrees of stability to generate PWMs codes. Experiments showed that the PBBM and PWMs features had higher robustness and reliability than the basic LBP methods. By selecting more discriminative bits from PBBM, the personalized discriminative bit map (PDBM) method was designed in [104]. The PDBM method contains more discriminatively information and fewer bits, so it has less computation complexity than PBBM. Sikarwar [105] discussed the application of the local directional pattern (LDP) method, which is the response of derivative directions among the local neighbors, for recognition, and the results confirmed that it helped to overcome the limitation of the LBP method. Meng *et al.* [106]

used the local directional code (LDC) method to represent the directional information hidden in the vein pattern and encoded them as a decimal number for recognition.

Histogram-based descriptor method, which can represent the overall distribution of finger-vein images, was also introduced for recognition. Lu *et al.* [107] used a fusion feature extraction method, which combined the histogram of competitive orientations (HCOs) and the histogram of competitive magnitudes-based LBP (HCMLBP), and the proposed HCOs and magnitudes (HCOM) method efficiently and adequately represented the finger-vein image. To effectively extract the rich orientation information, the pyramid histogram of the double competitive pattern (PHDCP) was designed in [108], where the orientation information was selected by the robust output of Gabor filters, and then, the column-partition-based pyramid histogram extraction method was used to encode the features. Compared with other local descriptors, the PHDCP contains more discriminative features.

The output of transform, which has information of magnitude, orientation, and phase in a specific scale and orientation, has been widely used in feature extraction [109], [110]. Yang *et al.* [109] computed the average absolute deviation (AAD) from the outputs of a set of Gabor filters for feature extraction, and the extracted features were then classified by a cosine similarity measure classifier. Another Gabor filter-based feature extraction scheme was used in [110], where the competitive coding scheme was used to extract features from the results of Gabor features. These Gabor filter-based features can extract local and global features and achieve good performance. Randon transform was also proposed in finger-vein extraction [111], [112]. Da Wu *et al.* [111] utilized Randon transform for feature extraction, and then, the artificial neural networks (ANNs), radial basis function (RBF) neural network, and probabilistic neural network (PNN) were applied for classification. Wang *et al.* [112] introduced a feature extraction scheme, where the Randon transform and SVD were proposed, to represent finger-vein image. The Randon transform is sensitive to image rotation and translation, which affects the recognition performance. Park [113] introduced a hybrid feature extraction algorithm, where the local features were represented by LBP and the global features were represented by wavelet transform. Then, the extracted features were classified by SVM and achieved good performance without segmenting accurate finger-vein regions.

The texture, shape, and other biometric traits which can describe the diversity of the vein patterns can be adapted for recognition. For example, the integrated responses of texture (IRT) features were proposed in [114] to detect the local and the global features, and experiments demonstrated that the IRT improved the recognition performance and resisted the effect of image rotation. To overcome the limitation of noise and low local consistency, superpixel features were adapted for feature representation. Liu *et al.* [115], [116] utilized the superpixel-level feature to obtain the base attributes and represent the finger-vein images, and the matching schemes were used to implement matching. Zhou *et al.* [117] explored the superpixel context feature (SPCF) for finger-vein recognition.

The SPCF was detected by comparing the superpixel with its surrounding neighbors. The SPCF method achieved better on two public finger-vein databases (HKPU and SDU) than the existing superpixel-based method. The soft biometric trait was explored in [118], and the extraction criteria of the soft biometric trait were proposed for feature extraction. Especially, the width of the phalangeal joint was introduced as a novel soft biometric trait and showed promising improvement of recognition performance.

The dimensionality reduction methods, such as principal component analysis (PCA), linear discriminant analysis (LDA), and entropy component analysis (ECA), were introduced to restore the main features and remove noises. Guan *et al.* [119] combined the advantages of the modular method and bidirectional weighted bidirectional 2-D PCA (BWB2-DPCA) with eigenvalue normalization and designed a bidirectional weighted modular bi-directional 2-D PCA (BWMB2-DPCA) algorithm to effectively represent the finger-vein images. A hybrid feature extraction method which consists of PCA and LDA was proposed in [120], while the PCA and LDA were used to retain main features and remove noises. The two-direction weighted horizontal and vertical 2-D LDA [(W2D)2LDA]-based algorithm was applied in [121], where the (W2D)2LDA method effectively extracted the main information and achieved better performance than the basic LDA method. The performances of PCA, kernel PCA (KPCA), and kernel ECA (KECA) were compared in [122] and [123], where the KPCA performed the best according to experiments. These dimensionality reduction methods are all based on mathematical knowledge and not optimized for finger-vein images. Thus, the performance of dimensionality reduction methods was greatly affected by the image quality or factors and generally was not very good when compared with other feature representation methods.

3) *Learning-Based Method*: The template-based method and representation-based method are all based on the prior knowledge of finger-vein and other related feature extraction methods. These handcrafted methods have achieved good results on one or some specific databases. However, these methods are not always working and susceptible to various factors that have been discussed above. Based on advanced knowledge of machine learning, these feature extraction methods can be further improved to achieve better results. The manifold learning method, which consists of PCA, locally linear embedding (LLE), and orthogonal neighborhood preserving projections (ONPPs), was proposed in [122] and robust enough against noise and deformation. The manifold learning method based on an anchor-based manifold binary pattern (AMBP) was present in [124], where the pixel difference vectors (PDVs) were extracted and dimension was reduced based on the goal of minimizing the quantization loss between the original real-valued vectors and learned binary codes. The proposed manifold learning-based method can preserve the manifold structure between the images.

To improve the performance of recognition and reduce computational complexity, Xi *et al.* [125] proposed a personalized subset features of pyramid histograms of gray, texture, and orientation gradients (PFS-PHGTG) methods. The abundant

information of finger-vein features was extracted by PHGTOG method, and the features from PHGTOG were trained by the least absolute shrinkage and selection operator (LASSO) method to further boost the recognition performance. The discriminative binary codes (DBC)s learning method was introduced in [126], where the binary templates that were transformed by relation matrix among subjects were used to describe vein features and SVMs were trained to obtain binary codes. The results showed that the DBC was more discriminative and had better performance than other binary code methods in the HKPU database and SDU database. Liu *et al.* [127] utilized a local discriminative feature learning method, which enhances the discriminative ability by learning a feature mapping. The multidirectional PDVs (MDPDVs) were extracted in [127], and then, a feature mapping, which aims to minimize the within-class variation of the local binary features and maximize the between-class variation of the local binary features, was learned to represent the MDPDVs. Experiments on the SDU and HKPU database verified the superior performance. Inspired by the bit-consistency phenomenon of LBP, Liu *et al.* [128] combined the constraints of intraclass sparse constraints with LDA criteria in the learning process to design a learned binary code for finger-vein recognition. Based on the constraints of quantization error minimizing, intraclass variance minimizing, and interclass variance maximizing, the proposed method represented the finger-vein subjects and results well on the SDU database and HKPU database. Su *et al.* [129] applied a binary hash code learning method to represent the finger-vein image in Hamming space. The nonlinear subspace coding (NSC) method was proposed in [129] to extract the features, and then, a binary hash codes learning method was designed to discriminatively represent the extracted binary codes. Experiments demonstrated the effectiveness and efficiency of the proposed binary hash coding learning method for large-scale finger-vein recognition. Yang *et al.* [130] introduced a finger-vein recognition framework, which combined the proposed indexing method and matching method to improve performance and efficiency. The extracted vein patterns were represented by an elliptical direction map as a feature for indexing. The similarity of features was measured by the grouped hamming distance in indexing and further weighted by the overlap degree of the corresponding vein patterns. The results verified the effectiveness of the proposed code indexing method with highly improved accuracy.

As an inner trait, finger veins are not easy to be copied, but the registration template may be stolen, which affects the security of the recognition system. A series of methods have been focused on protecting the templates and can be divided into two categories: biometric cryptosystems and feature transformation methods. The biometric cryptosystems combine biometric and cryptographic knowledge, while the feature transformation methods adopt transformation functions for security. Based on biometric encryption technology, Zhang and Fang [131] used fuzzy vault method and fuzzy extractor methods, respectively, to generate finger-vein templates for verification. However, both methods require more time cost because the generated template needs decryption before

recognition. Considering the location of vein intersection points, a quantification-based finger-vein template protection method based on the biometric encryption method is proposed in [27]. For feature transformation methods, Krivokuca and Marcel [132] introduced biohashing method, which required finger-vein images and user tokens for recognition. Shao *et al.* [133] designed a chaotic map-based finger-vein template protection method, which can ensure the safety of finger-vein templates and stability of recognition performance. In order to ensure the safety of the finger-vein template, Liu *et al.* [134] used deep-learning-based method to generate secure finger-vein template for verification. By converting the original template into a secure template, it is difficult to determine the corresponding relationship between the security template and the user, thereby protecting personal privacy.

The template-based methods and representation-based methods require researchers to have professional prior knowledge and be able to combine image processing technology, mathematical knowledge, biomedical knowledge, and other knowledge to design features that can reflect finger-vein information. The prior knowledge mastered by researchers will affect the quality of feature extraction, and the effect of subsequent feature matching and recognition often depends on the quality of feature extraction. Extracting the feature information that effectively reflects the research object is helpful to accurately realize finger-vein recognition. Therefore, the abovementioned finger-vein recognition method based on artificially designed features relies on effective feature extraction technology, and the feature extraction process with manual participation will add certain uncertainty to the recognition system. The method based on feature learning introduces the relevant knowledge of machine learning. Through learning, the feature values of finger veins can be extracted, which further improves the feature extraction method and achieves good results in the feature extraction of finger veins.

D. Matching

The matching step is the last step of finger-vein recognition. By measuring the similarity between the registered template and the input finger-vein image, the matching score is generated to discuss whether the input image is genuine or an imposter for the enrolled image. In finger-vein recognition, various types of matching methods, such as the matching-based method and classifier-based method, have been proposed to produce the matching score.

1) *Matching-Based Method*: Generally, the choice of a matching method depends on the result of the feature extraction method. For matching line-shaped patterns, template matching method is more appropriate [5], [34], [60], [66], [80], [83], [84], [89], [135], [136]. To overcome the pattern distortion, the robust template matching method, called “Miura match,” was proposed in [5], and only “ambiguous regions” around the veins were identified.

To improve the correct recognition accuracy, the phase-only-correlation (POC) method was proposed in [42], [43], [46], [88], [137], and [138] for finger-vein image matching. When two images are similar, the sharp peak can be

distinguished by the POC method, whereas there is no distinguishable peak when the images are not similar. Assuming that $M_i(x, y)$ and $M_j(x, y)$ represent two images (x and y are the pixels of the image), and $F_i(u, v)$ and $F_j(u, v)$ are (u and v are used to determine the frequency of the sine and cosine) their 2-D Discrete Fourier transform results, according to the Fourier transform [137], [138]

$$M_i(x, y) * M_j(x, y) \Leftrightarrow F_i(u, v) \overline{F_j(u, v)} \quad (1)$$

where the symbol $*$ represents a 2-D correlation operator, so the cross-phase spectrum is

$$R(u, v) = \frac{F_i(u, v) \overline{F_j(u, v)}}{\|F_i(u, v) \overline{F_j(u, v)}\|} = e^{j\theta(u, v)}. \quad (2)$$

Then, $r(x, y) = \text{IDFT}(R(u, v))$ is called the POC function or measure. The POC results of images from two different classes are near to zeros. The POC method is robust to noise and image shift, thus makes it very helpful for finger-vein image recognition [137], [138].

As the most commonly used distance methods, Euclidean distance [51], [53], [64], [107], [108], [139], [140], cosine distance [57], [81], [109], [117], [127], Manhattan distance [105], and chi-square distance [108] have been widely used to calculate similarity. For binary features, the matching score is usually estimated by Hamming distance [37], [52], [56], [76], [97], [99], [102], [103], [110], [141]. The formula is given as

$$\text{HD} = \frac{\|\text{codeA} \otimes \text{codeB}\|}{L}, \quad \text{HD} \in [0, 1] \quad (3)$$

where codeA and codeB are the extracted binary features of the template image and the input image. The symbol $\otimes$ represents Boolean exclusive-OR operator between the corresponding pair of bits, and L represents the length of extracted template binary code. Yang *et al.* [141] applied a hamming distance-based matching score fusion method for finger-vein recognition. The matching score was delivered by Hamming distance and then fused by a combination-based method to make the decision. The multi-instance matching score fusion could significantly improve recognition performance, according to the experiments. The modified Hausdorff distance [7], [41], which measured the similarity between two minutia point sets, was also proposed for finger-vein minutia image matching. Let $X = \{x_1, x_2, \dots, x_n\}$ represents minutia point set of input image, x_i represents a minutia point, $Y = \{y_1, y_2, \dots, y_n\}$ represents minutia point set of template image, and y_i is a minutia point. The modified Hausdorff distance is defined as follows:

$$\text{MHD} = \frac{1}{n} \sum_{x_i \in X} \|x_i - y_i\|, \quad \text{MHD} \in [0, 1]. \quad (4)$$

Chiu *et al.* [142] used the Bhattacharyya distance (BD), which measures the distribution of two discrete probability, to measure the similarity of two histogram features. The smaller the value of Bhattacharyya distance (BD), the greater the similarity score. The Bhattacharyya coefficient (BC) and BD are defined in (5) and (6), respectively, where $p(x)$ and

$q(x)$ represent the histogram features of two finger-vein image

$$\text{BC} = \sum_{i=0}^{255} \sqrt{p(x) \times q(x)}, \quad 0 \leq \text{BC} \leq 1 \quad (5)$$

$$\text{BD} = -\ln(\text{BC}), \quad 0 \leq \text{BD} \leq \infty. \quad (6)$$

2) *Classifier-Based Method*: The classifier-based method, especially the machine learning method, has advantages in solving complex pattern recognition problems and is effective in recognition. The SVM [100], [113], [114], [120], [143] is the most popular classifier compared to other classifiers. Kang *et al.* [143] adopted a hybrid matching strategy, which consists of SVM and Manhattan distance, for finger-vein recognition. The SVM, which performs better in high-dimensional vector classification, was used to identify primary biometric traits, and Manhattan distance was used for soft biometric trait matching. This hybrid method comprehensively considered the advantages of the two methods and achieved satisfactory performance. Compared with the neural network, the SVM took much less time for training and testing and had an equally good performance as adaptive neuro-fuzzy inference system (ANFIS) [120]. The extreme learning machine (ELM), which shows better generalization performance and faster learning speed, has also been introduced in finger-vein recognition. The feature component-based ELMs (FC-ELMs) were utilized in [144], and the FC-ELM consists of eight single ELMs. The outputs of each ELM, which trained with a block feature in a predefined direction, were combined to classify the input finger-vein images. Due to the less complex boundary required for each ELM, the proposed FC-ELMs worked stability and performed well. Yang *et al.* [145] introduced a metric learning method called large margin nearest neighbor (LMNN), which overcomes the shortage of K -nearest neighbor (KNN) classifier by learning a distance metric. Considering the unbalance of positive samples and negative samples, the synthetic minority oversampling technique (SMOTE) technology was used for data augmentation. The proposed KNN classifier using LMNN changed the sample distribution in the new metric space and thus improved the recognition performance.

For matching-based methods, “Miura match” is specially designed and has achieved good performance in image template matching. For feature-matching, the POC method works better than other distance-based method. Among the classifier-based methods, SVM method is the most frequently used method and has achieved good results.

E. Performance Analysis

For evaluation, researchers have designed four indicators, namely, FRR, FAR, EER, and accuracy. For ease of introduction, the classification of finger vein is simplified as a binary classification problem. When the classifier predicts the same results as the true class, the correct results are achieved, namely, true accept and true reject. The false accept and false reject are shown when the classifier gives opposite results to the true class.

False Rejection Rate: The FRR typically represents the number of false rejections (in terms of percentage). It does

TABLE IV
SUMMARY OF TEMPLATE-BASED FINGER-VEIN FEATURE EXTRACTION METHOD

Ref	Year	Feature Extraction	Matching/Classifying	Classes × Per fingers	Performance
[34]	2010	Wide line detector (WLD)	Template matching	10140 × 5	EER: 0.87%
[83]	2011	Region growth	Template matching	320 × 5	EER: 0.037%
[85]	2017	Enhanced maximum curvature & HOG descriptors	SVM	636 × 6	EER: 0.14%
[86]	2017	Anatomy structure analysis based vein extraction(ASAVE)	Elastic matching	312 × 6	EER:0.38%
				636 × 6	EER:1.39%
[87]	2013	Difference curvature & SIFT & shape and orientation features.	Generating fusion matching	426 × 10	EER:2.75%
[89]	2011	Mean curvature	Matched pixel ratio method	125 × 9	EER: 0.25%
[90]	2017	Random transform	Template matching	316 × 6	EER:1.03%
[91]	2021	Aggregation of Radon-like features	Conventional template-matching	312 × 6/12	EER:2.28%
				600 × 10	EER:0.77%
				492 × 12	EER:0.74%
				6180 × 6	EER:1.39%
[92]	2020	kernel fuzzy C-means (KFCM) algorithm	Template matching	312 × 6/12	EER:2.35%
[93]	2009	Minutiae points	Modified Hausdorff distance	50 × 10	EER:0.761%
[94]	2014	Local average intensity (LAI) & local intensity deviation (LID) & local extensive binary pattern (LEBP)	SVD-pairing matching	312 × 6	EER:5.01%
				136 × 30	EER:2.46%
[95]	2017	Tri-branch vein structure	Template matching	312 × 6	EER:0.74%
				636 × 6	EER:3.46%
[136]	2011	Vein location and direction coding	template matching	220 × 2	EER:0.44%

not necessarily indicate a flaw in the biometric system. The FRR measures the probability of false rejecting an attempt by a register when the matching score of the same finger vein is lower than a given threshold. It is defined as

$$FRR = \frac{\text{number of genuine rejects}}{\text{number of genuine attempts}} \quad (7)$$

False Acceptance Rate (FAR): The FAR is the number of times when the predicted results are in a false acceptance. The false acceptance occurs when the matching score of the different finger vein is greater than a given threshold, thus making it accepted by the system as the same finger vein. The FAR can be calculated as

$$FAR = \frac{\text{number of imposter accepts}}{\text{number of imposter attempts}} \quad (8)$$

Equal Error Rate: The EER is defined when the FAR and FRR are equal. If the threshold is increased, the FAR is decreased and FRR is increased. Similarly, the FAR is increased to evaluate recognition performance. The FRR is decreased when the threshold is reduced. In fact, the equal error rate can be obtained intuitively by adjusting the threshold, as shown in Fig. 10.

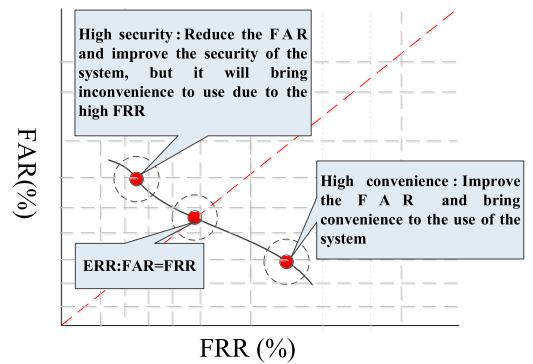


Fig. 10. Relationship of FRR, FAR, and EER.

Accuracy (Acc): The Acc, which represents the number of times of correct recognitions, is a classic indicator used

$$Acc = \frac{\text{number of correct predictions}}{\text{total number of predictions}} \quad (9)$$

Based on the discussion of feature extraction methods and matching methods, Table IV summarizes the template-based feature extraction method, Table V shows the represent-based

TABLE V
SUMMARY OF REPRESENT-BASED FINGER-VEIN FEATURE EXTRACTION METHOD

Ref	Year	Feature Extraction	Matching/Classifying	Classes $\times$ Per fingers	Performance
[97]	2015	LBP	Hamming distance	636 $\times$ 6	EER:47.30%
[99]	2011	LLBP	Hamming distance	204 $\times$ 10	EER:3.28%
[100]	2015	Local histogram of texture and magnitude	SVM	636 $\times$ 6	EER:0.36%(Right) EER:0.39%(Left)
[101]	2018	Local hybrid binary gradient contour	Score level fusion method	636 $\times$ 6	EER:0.007%
				492 $\times$ 12	EER:0.00002%
[102]	2012	PBBM	Hamming distance	106 $\times$ 14	EER:0.38%
[103]	2013	PWMs	Hamming distance	136 $\times$ 20	EER: 0.56%
[105]	2016	Local directional patterns	Manhattan distance	636 $\times$ 6	EER:14.3082% Acc:86.1635%
[106]	2012	Local directional code	Matching score	136 $\times$ 30	EER:1.02% Acc:100%
[107]	2017	Histogram of competitive orientations and magnitudes	Euclidean distance	600 $\times$ 10	EER:0.36%
[108]	2018	Pyramid histogram of double competitive pattern	Chi-square (C)/ Euclidean distance (E)	600 $\times$ 10	EER: 0.33% (E) EER:0.23% (C)
				360 $\times$ 4	EER: 2.91% (E) EER:2.36% (C)
[109]	2011	Gabor filter features	Cosine similarity	70 $\times$ 30	Acc: 98.097% FAR: 0.9523%
[110]	2012	Gabor filters based comparative competitive coding"	Modified Hamming distance	610 $\times$ 4	EER: 0.45%
[112]	2010	Radon Transform and SVD	Normalized distance	10 $\times$ 10	FAR: 2.0% FRR: 0.0%
[113]	2020	LBP & Wavelet transform	SVM	400 $\times$ 10	EER: 0.011%
[114]	2015	Integrated Responses of Texture (IRT) features	Chi-Square Distance method	636 $\times$ 6	EER:0.5%
[115]	2013	LBP and super-pixel level feature	Weighted average strategy matching score	136 $\times$ 30	Acc: 98.01% EER: 1.56%
[117]	2016	Superpixel context feature	Cosine distance	312 $\times$ 6	EER:0.75% Acc: 97.65%
				636 $\times$ 6	EER: 1.94% Acc: 92.71%

feature extraction method, and Table VI summarizes the learn-
ing feature-based feature extraction method.

IV. DEEP-LEARNING-BASED METHOD

Most of the current handcrafted features are affected by the feature extraction methods and matching strategy. For example, handcrafted features are sensitive to noise and do not perform well on low-quality images. Besides that, it is hard to set standard parameters for methods and takes lots of experiments to determine the suitable parameters for different databases. Moreover, the choice of matching strategy, after the feature extraction, can also affect the recognition performance. To overcome the limitation of the traditional handcrafted-based method, the deep learning method, which has replaced the handcrafted-based method in the field of computer vision,

has been introduced for finger-vein recognition. Inspired by human brain, deep learning methods, such as convolutional neural networks (CNNs), can be used to learn hierarchical representation/features from data. Compared with the traditional methods, a significant advantage of deep learning is that it can extract features, reduce data dimension, and classify in a network structure simultaneously. In addition, the deep learning methods can obtain good and stable recognition performance, regardless of the image quality and placement of the finger. Deep learning methods have been successfully used for image quality assessment, image restoration, feature extraction, and classification.

In image quality assessment, the deep-learning-based method has been applied in low-quality image prediction. Based on the assumption that low-quality image is false

TABLE VI
SUMMARY OF LEARNING FEATURE-BASED FINGER-VEIN FEATURE EXTRACTION METHOD

Ref	Year	Feature Extraction	Matching/Classifying	Classes $\times$ Per fingers	Performance
[124]	2019	Anchor-based manifold binary pattern & Histogram representation & WPCA	Cosine similarity	636 $\times$ 6	Acc: 98.71% EER: 1.09%
				312 $\times$ 6	Acc: 99.89% EER: 0.42%
[125]	2013	Personalized subset of features from Pyramid Histograms of Gray, Texture and Orientation Gradients	Euclidean distance	136 $\times$ 30	EER: 0.22% Acc: 98.90%
[126]	2017	Discriminative binary codes	Hamming distance	312 $\times$ 6	EER: 1.44%
[127]	2018	Discriminative binary descriptor	Cosine similarity	636 $\times$ 6	EER: 1.89%
				312 $\times$ 6	EER: 0.69%
[128]	2019	Personalized binary code	Cosine similarity	363 $\times$ 6	Acc: 99.26% EER: 0.68%
				312 $\times$ 6	Acc: 100% EER: 0.30%
[129]	2019	Binary Hash codes learning method	Hamming distance	312 $\times$ 6	EER: 6.55%
[130]	2019	Weighted vein code index method	Elastic matching	312 $\times$ 6	Acc: 99.68% EER: 3.33%

rejected, Qin and El-Yacoubi [147], [148] designed a CNN to extract features from binary images for quality assessment and achieves high performance in the public database. Zeng *et al.* [149] adopted a light-CNN-based image quality assessment method to distinguish whether it contains rich and stable vein patterns. The image was segmented into many blocks, and then, average results extracted from these block images were used to represent the image quality. The proposed light-CNN architecture not only decreases the computation time but also extracts robust features to represent finger-vein image quality. Compared with handcrafted descriptor-based methods, the deep-learning-based method can automatically learn robust features for image quality evaluation and achieve significantly better performance.

In image restoration, Jing *et al.* [150] utilized an autoencoder network to restore the contaminated image. The designed autoencoder network used hole convolution to increase the receptive field of convolution and achieved significant results in finger-vein restoration. The modified conditional generative adversarial network (conditional GAN) was proposed in [151] for vein image restoration. Taking advantage of the powerful learning and representation ability of GAN, factors that lead to image blurring are all taken into account, and experiments on two publicly database show outperformance in image restoration.

Based on its strong feature learning capabilities, the deep-learning-based method has achieved excellent performance in feature extraction and classification. Qin *et al.* [152] designed a deep learning model for vein pattern extraction and recovery. The probability of pixels belonging to vein pattern networks and background is predicted, segmented, and recovered by CNN. The proposed deep-learning-based method is robust to different databases and environment changes when compared

to traditional methods. Combined CNNs and long short-term memory (LSTM), the SCNN-LSTM was adopted in [153]. The outputs of SCNN-LSTM represented the vein patterns, and then, a probabilistic SVM (P-SVM) was used to predict the probability of pixels belonging to vein networks. In SCNN-LSTM, CNN learned the robust features of finger-vein image and LSTM stored the complex spatial relationship of vein patterns. Taking advantage of the ability of autoencoder in dimension reduction and feature extraction, Hou and Yan [154], [155] presented convolutional autoencoder-based methods to learn the high-level features. Experimental results proved that the proposed method achieved better performance in learning features than the traditional dimension reduction method. Jalilian and Uhl [156] evaluated the performance of three different classical autoencoder architectures in finger-vein pattern extraction. The training and configuration settings of autoencoder architectures, U-net, RefineNet, and SegNet, were explored, and the experiments on the SDU database and UTFVP database accentuated the fact that autoencoder architectures with automatically generated labels improved the performance of pattern extraction. Yang *et al.* [157] designed a cycle GAN (Cycle-GAN)-based method termed finger vein generative adversarial network (FV-GAN) for pattern extraction. The FV-GAN was designed to represent vein patterns directly without prior knowledge. Song *et al.* [158] used the composite image as the input of a CNN network for feature extraction, followed by a shift matching method for matching. The proposed method with composite image and shift matching method has strong robustness to noise and misalignment, according to experiments. Zeng *et al.* [159] designed an end-to-end model, which integrated the fully CNN (FCNN) with conditional random field (CRF) to extract vein pattern network for finger-vein verification.

Ahmad Radzi *et al.* [160] applied a simple four-layer CNN model for feature extraction. The tenfold cross-validation method was adopted to reveal the excellent performance of this model. Using the ability of feature learning, the autoencoder networks were utilized in [161] to represent finger-vein images. This auto feature extraction and learning process achieved high accuracy in the SDU database. Based on the typical structure of AlexNet, Meng *et al.* [162] proposed a CNN model, which includes five convolutional layers and three fully connected layers, to extract feature vectors from finger-vein images, and Euclidean distance between two vectors was calculated to measure the similarity. Das *et al.* [163] used four public finger-vein databases that are in different image quality levels to evaluate and verify the effectiveness of CNN. The results of experiments confirmed that the CNN method was robust to image quality in feature extraction and classification. Tao *et al.* [164] used mask region CNN (Mask-RCNN) to obtain the accurate ROI image and then constructed a softmax-based decision algorithm for verification. However, the weakness of these discussed methods is that the design of the CNN structure needs to be determined through lots of experiments. Besides, the network training needs a large number of images for the desired performance, which is impractical. Moreover, there are amount of trainable parameters in CNN, so the training time is high and not suitable for real-time recognition.

To achieve better performance, the pretrained network has been adopted in finger-vein recognition. The usage of pretrained models can reduce the training time complexity, but the extracted features are not compact enough. Hong *et al.* [165] used a pretrained VGG-16 model after fine-tuning for finger-vein recognition. Then, some hybrid feature extraction methods that combined the handcrafted-based feature extraction methods and the deep-learning-based methods have been proposed to achieve higher performance. As a combination of pretrained networks and artificial features, the pretrained neural network was used to extract high-dimensional features, and then, the hand-crafted method was utilized to further encode high-dimensional features into compact and discriminative feature vectors. For example, the pretrained model was used in [166] to design a CNN-based local descriptor named CNN competitive order (CNN-CO). The outputs of selected CNN filters were used to generate CNN-CO, which can represent finger-vein images discriminatively and performed better than other existing local descriptors on two public databases. Li *et al.* [167] introduced a local coding-based CNN (LC-CNN), where the local coding convolution was proposed in pretrained CNN, and then, extracted features were put into SVM for classification. The co-occurrence features among pretrained convolutional filters were modeled in multiweighted co-occurrence descriptor encoding (MWCDE) to represent finger-vein image in [168], and then, these high-order descriptors were embedded into the compact feature vectors. The potential in pretrained CNN can be fully exploited by combining the descriptors with feature maps in high-level layers of CNN, where the existence of similar patterns can be observed and the problem of redundancy can be resolved. Experiments demonstrated the discrimination and generalization ability of the pretrained model-based finger-vein

recognition method. However, these pretrained models usually have redundant parameters, which are not trained specially for finger-vein recognition, and are not conducive to miniaturization for real-time recognition.

Some technologies, such as data augmentation and metric learning methods, have been introduced in deep learning methods to improve the feature learning ability when the amount of finger-vein database is limited. For example, a GAN model, named FCGAN, was utilized in [169] for finger-vein image augmentation. The experiments revealed that generated finger-vein images were helpful in classification and achieved higher performance than the classic sample augmentation methods.

Metric learning, which is used to measure the samples' distance, has been proposed in finger-vein recognition to perform a robust classification when training data are not enough. The purpose of metric learning is to reduce intraclass distance and increase interclass distance through training. Huang *et al.* [170] used contrastive loss to train CNN model for finger-vein recognition, and Tang *et al.* [171] modified the contrastive loss to extract more discriminative features. In contrastive loss, intraclass samples and interclass samples were paired and generated as input data. The training guidance of this method is maximizing the distance of interclass pairs and minimizing the distance of intraclass ones. Fang *et al.* [172] designed a two-stream model that integrated the original image and the mini-ROI image to solve the problem of displacement and the training data limitation. Based on the features extracted by the two-stream network, the recognition task was accomplished and achieved the state-of-the-art with an acceptable computation cost, making it viable in practice. Hu *et al.* [173] proposed a large margin softmax loss-based model, which used a pretrained model and augmentation strategies. Then, a template-like matching strategy was designed for recognition. The introduction of a pretrained model reduced the time required for network design, augmentation strategies improved robustness to rotation and translation, and large margin softmax loss increased the expressiveness. Zhao *et al.* [174] used a center loss-based metric learning model, which makes each class easier to be distinguished. In the center loss, the Euclidean distance between each data and its corresponding category center is measured and constantly updated during training and paired requirement for input data is not necessary.

Using cosine distance instead of Euclidean distance, Hou and Yan [175] proposed arccosine center loss to improve the discriminative ability. The main purpose of arccosine center loss is to reduce intraclass distance and increase the interclass distance through network model training. Experimental results illustrated the effectiveness and efficiency of the arccosine center loss for finger-vein verification. Combined cosine softmax loss and triplet loss, Ou *et al.* [176] designed a joint finger-vein verification method, and an interclass data augmentation method was also introduced to improve the generalization ability of learned features. Through the metric learning module, the trained network can learn features with interclass and intraclass information, which improves the ability of finger-vein recognition.

Moreover, other types of deep-learning-based methods have been used in finger-vein recognition. The PCA network

(PCANet), which combines cascaded PCA, binary hashing, and block-wise histogram, was utilized in [177]. The proposed PCANet consists of two parts, the convolutional parts which learn the features by filter banks of PCA and the output part which converts the features to binary hashing and blockwise histogram for classification. Using a filter generation method that computes the correlation between grayscale image and vein pattern image by canonical correlation analysis (CCA) method, the PCANet-based method achieved higher accuracy in three public databases, USM, SDU, and THU-FVDT2, than other methods. Zhang *et al.* [178] combined CNN with a Gabor filter in the field of biometrics. The adopted method was able to learn the parameters of Gabor filters adaptively and performs better than other Gabor-based methods. Li *et al.* [179] proposed an improved ChebyNet graph CNN to extract graph-level features of weighted graph and achieved better performance in recognition accuracy and matching efficiency.

As we all know, the training data of finger-vein images have an immeasurable impact on the performance of finger-vein recognition. Facing with the lack of training data, some researchers attempt to apply prior knowledge learned on ImageNet to target tasks by transfer learning to reduce the amount of training data impact on model training. However, the quality of the effect is related to the relevance between source knowledge and the target task. Considering the difference in the distribution of the finger-vein data and the ImageNet data, the effectiveness of the prior knowledge applied in finger-vein recognition will be greatly reduced. Recently, GANs were successfully applied in computer vision, such as image style transfer and image superresolution. Some researchers have applied GANs for data augmentation to expand the sample of finger veins.

In order to measure the performance of deep learning methods, there are two experimental protocols, closed-set protocol and open-set protocol. The closed-set protocol uses partial samples of all of the classes in the database for training and the remaining samples for testing. Most deep-learning-based methods are measured under the closed-set protocol, which is more suitable for those scenarios without new user registration. However, in actual scenarios, the pretrained model may need to work well on untrained classes. Therefore, the open-set protocol, only a part of classes is used for training and the remaining classes are only used for testing, is proposed for measurement. In general, the results under closed-set protocol can be achieved through classifying, while the results under open-set protocol depend on the matching of the extracted features. It is noteworthy that compared with the close-set protocol, the recognition challenge of the open-set protocol is higher, and it can better illustrate the generalization ability of the model. Table VII summarizes the deep-learning-based methods on the closed-set protocol, and Table VIII summarizes the deep-learning-based methods on the open-set protocol.

V. DISCUSSION AND SUGGESTION FOR FUTURE WORK

Finger-vein recognition is a promising biometric technology due to its high stability and safety, as finger vein is located under the skin. A lot of methods have been used for finger-vein recognition. However, there still exist three challenges.

First and foremost, the challenge is how to obtain a high-quality image. The quality and contrast of the image are usually important factors affecting finger-vein recognition performance. Currently, the top-light illumination-based capture device is widely used in the laboratory to obtain high-contrast finger-vein patterns, while the side-light illumination-based capture device is often used in the business scenario, due to its convenience. Hence, the optimized design of the capture device and appropriate image preprocessing method play important roles to obtain high-quality images. The existing image preprocessing methods help to improve the recognition performance, but it is hard to select the proper model to represent and improve the image quality. Many image quality evaluation, image enhancement, and restoration methods have been applied to overcome the image quality problems of image blurring, nonuniform illumination, low contrast, and thickness of fat, but it has not yet reached the level that can be widely used. Therefore, it is necessary to introduce more powerful deep-learning-based methods in image preprocessing. Moreover, some of the image preprocessing methods need longer processing time, thus makes it not practical in real-life applications.

The second challenge is how to extract features effectively and then design a strong and fast matching strategy. The handcrafted-based methods, such as template-based method, representation-based method, and learning feature-based method, are susceptible to external factors of image quality, finger movement, and misalignment. The performance of the existing handcrafted-based methods depends on how the image preprocessing method is applied. The template-based method may be robust on public databases but not suitable in large-scale and real-time recognition. Representation-based methods and learning feature-based methods powerfully boost the performance, but they are not guaranteed when they captured images with high intraclass variation and low interclass variation. In fact, it is hard to design a feature extraction method that can efficiently extract discriminative features and find a suitable classifier with high performance and fast speed for handcrafted-based methods. The deep-learning-based method, which automatically learns robust attributes for feature representation and classification, leads to a significant improvement. Compared to the handcrafted-based method, the deep-learning-based method achieves higher performance but needs longer processing time. Therefore, it is necessary to introduce lightweight deep-learning-based method with powerful discriminative ability.

In order to extract more representative features and improve the recognition performance, the multimodal biometric methods have also been introduced into finger-vein recognition. Wan *et al.* [180] used the CNN to simultaneously recognize both finger veins and finger shapes, where high-level feature can be abundantly extracted by convolution operation. A fusion of iris, finger vein, and face biometric model was proposed in [181], where the CNN was used for feature extraction. Then, the feature- and score-level fusion approaches are applied to identify the identity of the individual. To overcome the spoof attacks, multimodal biometric authentication system based on deep fusion of electrocardiogram (ECG) and finger vein was utilized in [182], and experimental results

TABLE VII
DEEP-LEARNING-BASED FINGER-VEIN RECOGNITION METHOD ON CLOSED-SET PROTOCOL

Ref.	Year	Deep learning method	Division	Database	classes	Performance
[154]	2018	Auto-encoder & CNN	9:1	USM*	492	Acc: 99.49% EER: 0.16%
				SDU**	636	Acc: 83.02% EER: 6.28%
[155]	2020	Auto-encoder & SVM	9:1	USM*	492	Acc: 99.95% EER: 0.12%
				SDU**	636	Acc: 99.78% EER: 0.21%
[162]	2017	CNN	39:30:6	Proprietary	64	Acc: 99.4% EER: 0.21%
[166]	2019	CNN competitive order (CNN-CO) & Euclidean distance	5:5	MMCBNU	600	EER: 0.74%
			3:3	SDU	636	EER: 2.37%
[167]	2019	Weighted symmetric graph structure CNN & SVM	7:3	Proprietary**	500	Acc: 100% EER: 0.21%
				SDU**	636	Acc: 98.78% EER: 2.82%
[168]	2020	Multi-weighted co-occurrence descriptor encoding & Large margin distribution machine	6:6***	USM	492	EER: 0.307%
[169]	2019	GAN for augmentation & CNN for classification	6:6***	HKPU	210	EER: 0.52%
[172]	2018	Two-stream convolutional network & SVM	5:5	MMCBUN	600	EER: 0.10%
			5:1	SDU	636	EER: 0.47%
[173]	2018	Pre-trained VGGFace-Net & Large margin-Softmax loss	5:5	MMCBNU	600	EER: 0.04%
			6:6	USM	492	EER: 0.06%
			3:3	SDU	636	EER: 0.46%
[174]	2020	CNN with center loss	8:2	MMCBNU	600	Acc: 99.05% EER: 0.503%
			10:2	USM	636	Acc: 97.95% EER: 1.07%
[175]	2021	CNN with arccosine center loss	6:6	USM	492	Acc: 99.93% EER: 0.03%
			3:3	SDU	636	Acc: 98.51% EER: 0.72%
			6:6	HKPU	210	Acc: 99.60% EER: 0.55%
			3:3	GERWIN	1386	Acc: 99.20%
						EER: 1.59%
[178]	2019	Gabor convolution network	8:2	MMCBNU	600	EER: 0.11%
			8:4	USM	492	EER: 0.57%
			4:2	SDU	636	EER: 1.09%
			6:6***	HKPU	210	EER: 1.67%

* 3 images from each session were used as the training set and the remaining 3 from each session as the testing set.

** The size of the database was increased by 10 times.

*** The images from session 1 were used for training, and images from session 2 for testing

showed its effectiveness in performing secure authentication. iris, palm vein, and finger vein for multibiometric recognition, Zhou *et al.* [183] presented a novel hybrid fusion model of which included an improved feature fusion algorithm and a

TABLE VIII
DEEP-LEARNING-BASED FINGER-VEIN RECOGNITION METHOD ON OPEN-SET PROTOCOL

Ref	Year	Deep learning method	Division	Database	Performance
[151]	2020	Conditional GAN for restoring & CNN for classification	156:156*	HKPU **	EER: 1.814%
			318:318*	SDU	EER: 3.934%
[152]	2017	CNN based vein pattern recovery & CNN based feature extraction & matching	55:50:105	HKPU	EER: 2.70%
			123:123:246	USM	EER: 1.42%
[157]	2019	CycleGAN for finger-vein extraction & matching	300:40:270	THUFVDT2	EER: 1.12%
			300:40:636***	SDU	EER: 0.94%
[158]	2019	Composite image & Data Augmentation & CNN& shift matching	156:156*	HKPU**	EER: 0.33%
			318:318*	SDU	EER: 2.35%
[161]	2015	Sparse Auto-encoder	500:100	SDU	EER: 0.70%
[165]	2017	Pre-trained model of VGG Net-16	60:60*	Proprietary	EER: 0.396%
			98:98*	Proprietary	EER: 1.275%
			318:318*	SDU	EER: 3.906%
[169]	2019	GAN for augmentation & CNN for classification	210:636****	SDU	EER: 0.87%
[171]	2019	Siamese network with proposed MC loss	362:58:180	MMCBNU	EER: 0.17%
			296:48:148	USM	EER: 0.50%
			382:62:192	SDU	EER: 0.87%
			362:58:180	Proprietary	EER: 2.04%
[173]	2018	Pre-trained VGGFace-Net & Large margin-Softmax loss	300:300	MMCBNU	EER:0.30%
			246:246	USM	EER:0.76%
			318:318	SDU	EER:1.20%
[175]	2021	CNN with arccosine center loss	246:246	USM	EER: 0.25%
			318:318	SDU	EER: 1.53%
			105:105	HKPU	EER: 1.30%
			693:693	GERWIN	EER: 3.53%
[176]	2021	Inter-class data augmentation with cosine softmax loss and triplet loss	246:246	USM	EER: 0.48%
			300:300	MMCBNU	EER: 0.21%
		105:207***** HKPU			EER: 1.90%

* The two-fold cross-validation method is used for evaluation.

** Only the images of Session 1 were used in the experiments.

*** The images from the THUFVDT2 database were used for training and validation.

**** The images from the HKPU database were used for training and images from the SDU database for testing.

***** The images from last 51 subjects which have one session of data were used as extra test data.

novel weighting vote strategy and provided high precision and strong robustness.

The advantage of multimodal biometric system is that it fuses information from multiple modalities to provide higher security than most of existing unimodal biometric systems and is vulnerable to attacks. However, since acquiring different types of physiological features require different acquisition equipment and the fusion mechanism and method of different physiological features have not been clarified, it is difficult for multimodal biometric identification devices to be widely used.

The third challenge is related to the limited size of the existing database. Most of the existing methods perform well on small databases, but no guarantees in large databases.

Moreover, the promising performance of deep-learning-based methods also needs a large-scale finger-vein database for training. Additionally, the existing finger-vein databases are only collected by a single capture device, and the variation of the capture device is not taken into account. Therefore, the evaluation of the existing image preprocessing method and feature extraction method on these public databases ignores the impact of different capture devices. Hence, for future work, a large-scale standard finger-vein database collected by different capture devices is needed for existing methods' evaluation.

These challenges can be partially solved by introducing new powerful finger-vein recognition methods, and the

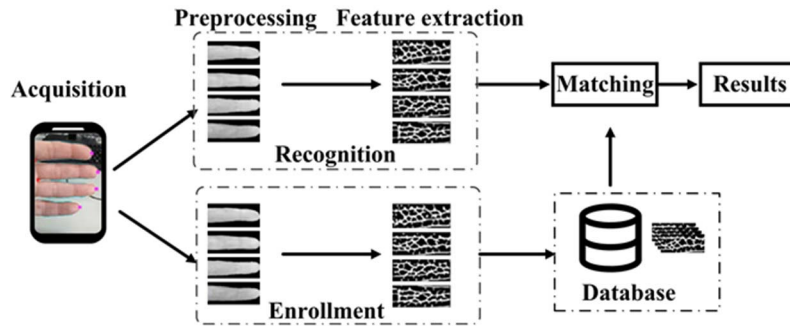


Fig. 11. Framework of visible-light-based finger-vein recognition system.

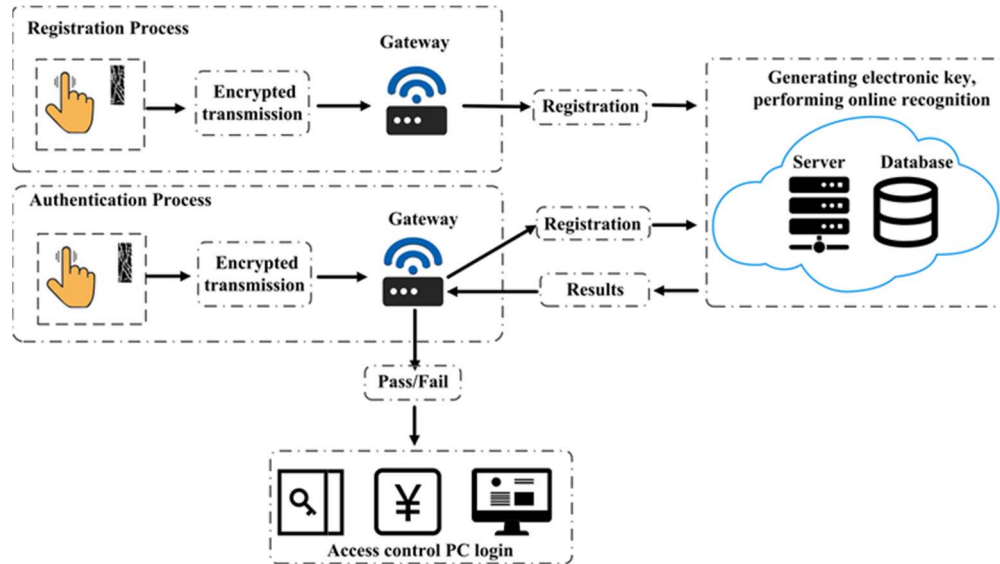


Fig. 12. Framework of online finger-vein recognition system.

improvement needs considerable advances in techniques. In fact, most of the current systems for finger-vein recognition rely on single-view 2-D images acquired from only one side and exist two limitations: the captured images have limited vein pattern information and the images captured from the same subjects are affected by finger position and rotation in contact-free mode. Both of these limitations have serious impacts on system performance. To provide a fundamental solution, the 3-D finger-vein recognition system has been proposed. With the help of 3-D reconstruction, most feature information, including all vein pattern information and 3-D geometric features of the finger, can be captured and then extracted for finger-vein recognition. Based on the knowledge obtained by cameras, Bunda [184] adopted the point cloud reconstruction method to build a 3-D model of the vein pattern. The full-view 3-D finger-vein recognition system, including 3-D finger-vein reconstruction, feature extraction, and matching, was proposed in [185]. A three-camera acquisition technology with three cameras was designed to simultaneously capture all finger-vein patterns of one finger, the new 3-D finger-vein reconstruction method was utilized to generate 3-D finger-vein structure, and a lightweight CNN method was introduced for recognition. Zhan *et al.* [186] demonstrated a

photoacoustic tomography-based 3-D finger-vein biometric verification system, which can provide robust imaging of the finger-vein network and obtain high performance. Compared with the traditional single-view 2-D model, the 3-D finger-vein recognition system achieved higher performance and showed greater potential in solving the misalignment problem. It is foreseeable that the 3-D-based finger-vein recognition method will become a research focus of the next generation of finger-vein recognition systems.

With its excellent security, a finger-vein recognition system has been used in many situations, including personal computer login, attendance management, bank automated teller machines (ATMs), and access control systems. However, the difficulty of the acquisition conditions, the complexity of the algorithm, and the high cost make this technology not widely used. Specifically, the finger-vein recognition system needs further research in some aspects to expand its scope of application.

Design of the Capture Device: Compared with fingerprint recognition systems, finger-vein recognition systems generally require larger acquisition equipment and more stringent acquisition conditions. As discussed above, the size of finger-vein acquisition equipment is restricted by current image

acquisition methods. Although in a commercial application, with the help of light refraction, GERWIN and other companies have designed compact finger-vein acquisition module, and the miniaturization of acquisition device is still a research hot spot. Limited by the current image acquisition method, it is very difficult to further reduce the size of the capture device. New finger-vein image acquisition methods, such as the bottom-light illumination method or visible-light method, need to be introduced into the design of finger-vein acquisition equipment. For example, Hitachi has designed a visible-light-based finger-vein recognition system that can be built into a mobile device. As shown in Fig. 11, the proposed system includes three key technologies: 1) background elimination and finger position correction; 2) vein pattern extraction; and 3) recognition based on multiple fingers. This visible-light-based finger-vein recognition enables it to be more widely used like fingerprint recognition. However, this system puts forward higher requirements for finger-vein image processing and feature extraction algorithm.

Type of Image Acquisition: As discussed above, the current finger-vein recognition system uses a 2-D image of a single finger for recognition. Limited by acquisition methods and equipment, 2-D finger-vein images have limited vein pattern information and are easily affected by finger movement and rotation. Therefore, it is meaningful to collect multiple fingers at a time or use 3-D finger-vein images for recognition. Multiple fingers and 3-D images can provide more finger-vein information, thereby improving the recognition performance and decreasing misrecognition.

Potential of the Recognition System: The advantages of finger-vein technology (unique, liveness, permanent, unforgeable, etc.) enable it to be used as public biometric infrastructure (PBI) for online authentication, instead of the password, smartcard fingerprints, faces, or other security technologies. Considering that the current finger-vein recognition systems are mostly applied in PC login and access control systems, it is very meaningful to study online secure biometric systems. As shown in Fig. 12, the online finger-vein recognition system designed by GERWIN Ltd. consists of a finger-vein capture device, server, gateway, and communication protocol. The finger-vein image obtained by the capture device is uploaded to the server for authentication through a communication protocol, and then, the server returns the authentication result. Different from the existing locally deployed finger-vein recognition system, the online system is more flexible and has the potential to attract more users to services, such as automatic payment solutions, self-check service, safety guarantee, and other situations requiring high security.

VI. CONCLUSION

This article comprehensively reviews most existing finger-vein recognition systems, including image acquisition, image preprocessing, feature extraction, and matching. Most of the existing publicly available finger-vein databases have also been introduced. First, the advantages and disadvantages of the existing image acquisition methods are discussed, and some academics and commercial devices are introduced. Then, in image preprocessing, image quality assessment,

image restoration, ROI extraction, and image enhancement, and normalization methods are discussed, and case studies on public database are used for presentation. Moreover, the handcrafted-based feature extraction methods are classified into three groups (i.e., template-based method, representation-based method, and learning feature-based method) and compared through performance analysis on public databases. Besides, the deep-learning-based methods are reviewed and compared with handcrafted-based methods. The performance on public finger-vein databases indicates that the deep-learning-based methods have significant improvement over traditional methods.

The challenges and prospects of the research are also discussed in this review. In particular, well-designed capture devices and image preprocessing methods are necessary for image quality improvement, and a large-scale database is also needed for evaluation. Moreover, the deep-learning-based methods efficiently improve the recognition performance in the application. Besides, the 3-D finger-vein recognition system shows great potential in the field of finger-vein biometric recognition. Research should focus on solving the existing problems and propose new more powerful methods for finger-vein recognition, such as the security of finger-vein recognition system based on deep learning, as the presentation attack research in the finger-vein recognition system has not been explored in scientific research.

REFERENCES

- [1] G. Betta *et al.*, "Face-based recognition techniques: Proposals for the metrological characterization of global and feature-based approaches," *Meas. Sci. Technol.*, vol. 22, no. 12, 2011, Art. no. 124005.
- [2] J. Soldera, C. T. J. Dodson, and J. Scharcanski, "Face recognition based on texture information and geodesic distance approximations between multivariate normal distributions," *Meas. Sci. Technol.*, vol. 29, no. 11, Nov. 2018, Art. no. 114001.
- [3] F. Wang and J. Han, "Information fusion in personal biometric authentication based on the iris pattern," *Meas. Sci. Technol.*, vol. 20, no. 4, 2009, Art. no. 045501.
- [4] T. Yanagawa, S. Aoki, and T. Oyama, "Diversity of human finger vein patterns and its application to personal identification," *Bull. Inform. Cybern.*, vol. 41, pp. 1–9, Mar. 2009.
- [5] N. Miura, A. Nagasaka, and T. Miyatake, "Feature extraction of finger-vein patterns based on repeated line tracking and its application to personal identification," *Mach. Vis. Appl.*, vol. 15, no. 4, pp. 194–203, 2004.
- [6] K. Syazana-Itqan, A. R. Syafeeza, N. M. Saad, N. A. Hamid, and W. H. B. M. Saad, "A review of finger-vein biometrics identification approaches," *Indian J. Sci. Technol.*, vol. 9, no. 32, pp. 1–9, 2016.
- [7] E. C. Lee and K. R. Park, "Restoration method of skin scattering blurred vein image for finger vein recognition," *Electron. Lett.*, vol. 45, no. 21, pp. 1074–1076, 2009.
- [8] C. Kauba, A. Uhl, and M. Drahansk, "Robustness of finger-vein recognition," in *Hand-Based Biometrics: Methods and Technology*. The Institution of Engineering and Technology, 2018, pp. 193–216.
- [9] C. Kauba and A. Uhl, "Sensor ageing impact on finger-vein recognition," in *Proc. Int. Conf. Biometrics (ICB)*, May 2015, pp. 113–120.
- [10] V. Ablinger, C. Zenz, J. Hammerle-Uhl, and A. Uhl, "Compression standards in finger vein recognition," in *Proc. Int. Conf. Biometrics (ICB)*, Jun. 2016, pp. 1–7.
- [11] B. Prommegger, C. Kauba, and A. Uhl, "Longitudinal finger rotation in finger-vein recognition," in *Proc. Int. Conf. Biometrics Special Interest Group (BIOSIG)*, Sep. 2018, pp. 1–8.
- [12] M. S. Al-Tamimi, "H. A survey on the vein biometric recognition systems: Trends and challenges," *J. Theor. Appl. Inf. Technol.*, vol. 97, no. 2, pp. 551–568, 2019.
- [13] K. Shaheed, H. Liu, G. Yang, I. Qureshi, J. Gou, and Y. Yin, "A systematic review of finger vein recognition techniques," *Information*, vol. 9, no. 9, p. 213, Aug. 2018.

- [14] R. Dev and R. Khanam, "Review on finger vein feature extraction methods," in *Proc. Int. Conf. Comput., Commun. Autom. (ICCCA)*, May 2017, pp. 1209–1213.
- [15] A. Kumar and Y. Zhou, "Human identification using finger images," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2228–2244, Apr. 2012.
- [16] Y. Yin, L. Liu, and X. Sun, "SDUMLA-HMT: A multimodal biometric database," in *Proc. Chin. Conf. Biometric Recognit.*, 2011, pp. 260–268.
- [17] M. S. M. Asaari, S. A. Suandi, and B. A. Rosdi, "Fusion of band limited phase only correlation and width centroid contour distance for finger based biometrics," *Expert Syst. Appl.*, vol. 41, no. 7, pp. 3367–3382, Jun. 2014.
- [18] Y. Lu, S. J. Xie, S. Yoon, Z. Wang, and D. S. Park, "An available database for the research of finger vein recognition," in *Proc. 6th Int. Congr. Image Signal Process. (CISP)*, vol. 1, Dec. 2013, pp. 410–415.
- [19] B. T. Ton and R. N. J. Veldhuis, "A high quality finger vascular pattern dataset collected using a custom designed capturing device," in *Proc. Int. Conf. Biometrics (ICB)*, Jun. 2013, pp. 1–5.
- [20] P. Tome, M. Vanoni, and S. Marcel, "On the vulnerability of finger vein recognition to spoofing," in *Proc. Int. Conf. Biometrics Special Interest Group (BIOSIG)*, Sep. 2014, pp. 1–10.
- [21] M. Vanoni, P. Tome, L. El Shafey, and S. Marcel, "Cross-database evaluation using an open finger vein sensor," in *Proc. IEEE Workshop Biometric Meas. Syst. Secur. Med. Appl. (BIOMS) Proc.*, Oct. 2014, pp. 30–35.
- [22] W. Yang, X. Yu, and Q. Liao, "Personal authentication using finger vein pattern and finger-dorsa texture fusion," in *Proc. 17th ACM Int. Conf. Multimedia (MM)*, 2009, pp. 905–908.
- [23] C. Kauba, B. Prommegger, and A. Uhl, "Combined fully contactless finger and hand vein capturing device with a corresponding dataset," *Sensors*, vol. 19, no. 22, p. 5014, Nov. 2019.
- [24] A. Lula, A. Savoia, and G. Caliano, "3D ultrasound palm vein pattern for biometric recognition," in *Proc. IEEE Int. Ultrason. Symp.*, Oct. 2012, pp. 1–4.
- [25] J. Lee *et al.*, "Imaging of the finger vein and blood flow for anti-spoofing authentication using a laser and a MEMS scanner," *Sensors*, vol. 17, no. 4, p. 925, Apr. 2017.
- [26] J. Lee *et al.*, "Performance comparison of illumination methods for finger-vein imaging and liveness detection," *Microsyst. Technol.*, vol. 24, no. 12, pp. 4955–4964, Dec. 2018.
- [27] J. Chavez-Galaviz *et al.*, "Embedded biometric cryptosystem based on finger vein patterns," in *Proc. 12th Int. Conf. Elect. Eng., Comput. Sci. Autom. Control (CCE)*, Oct. 2015, pp. 1–6.
- [28] Z. Liu and S. Song, "An embedded real-time finger-vein recognition system for mobile devices," *IEEE Trans. Consum. Electron.*, vol. 58, no. 2, pp. 522–527, May 2012.
- [29] S. Nandhini and D. Shyam, "Real-time finger-vein recognition for security and monitoring system," *Appl. Mech. Mater.*, vol. 550, pp. 194–203, May 2014.
- [30] S. Bazrafkan, T. Nedelcu, C. Costache, and P. Corcoran, "Finger vein biometric: Smartphone footprint prototype with vein map extraction using computational imaging techniques," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2016, pp. 512–513.
- [31] M. M. S. Ibrahim, F. S. M. Al Naimy, S. S. Amma, and L. Rajaji, "Biometric recognition for safe transaction using vein authentication system," in *Proc. IET Chennai 3rd Int. Conf. Sustain. Energy Intell. Syst. (SEISCON)*, 2012, pp. 77–82.
- [32] M. Waluś, K. Bernacki, and J. Konopacki, "Impact of NIR wavelength lighting in image acquisition on finger vein biometric system effectiveness," *Opto-Electron. Rev.*, vol. 25, no. 4, pp. 263–268, Dec. 2017.
- [33] R. Ramachandra, K. B. Raja, S. K. Venkatesh, and C. Busch, "Design and development of low-cost sensor to capture ventral and dorsal finger vein for biometric authentication," *IEEE Sensors J.*, vol. 19, no. 15, pp. 6102–6111, Aug. 2019.
- [34] B. Huang, Y. Dai, R. Li, D. Tang, and W. Li, "Finger-vein authentication based on wide line detector and pattern normalization," in *Proc. 20th Int. Conf. Pattern Recognit.*, Aug. 2010, pp. 1269–1272.
- [35] H. Qin, Z. Chen, and X. He, "Finger-vein image quality evaluation based on the representation of grayscale and binary image," *Multimedia Tools Appl.*, vol. 77, no. 2, pp. 2505–2527, Jan. 2018.
- [36] H. Ma, F. P. Cui, and P. Oluwatoyin, "A non-contact finger vein image quality assessment method," *Appl. Mech. Mater.*, vols. 239–240, pp. 986–989, Dec. 2012.
- [37] C.-H. Hsia, "New verification strategy for finger-vein recognition system," *IEEE Sensors J.*, vol. 18, no. 2, pp. 790–797, Jan. 2018.
- [38] L. Zhou, G. Yang, L. Yang, Y. Yin, and Y. Li, "Finger vein image quality evaluation based on support vector regression," *Int. J. Signal Process., Image Process. Pattern Recognit.*, vol. 8, no. 8, pp. 211–222, Aug. 2015.
- [39] L. Yang, G. Yang, Y. Yin, and R. Xiao, "Finger vein image quality evaluation using support vector machines," *Opt. Eng.*, vol. 52, no. 2, Feb. 2013, Art. no. 027003.
- [40] J. Yang and J. Wang, "Finger-vein image restoration considering skin layer structure," in *Proc. Int. Conf. Hand-Based Biometrics*, Nov. 2011, pp. 188–192.
- [41] E. C. Lee and K. R. Park, "Image restoration of skin scattering and optical blurring for finger vein recognition," *Opt. Lasers Eng.*, vol. 49, no. 7, pp. 816–828, Jul. 2011.
- [42] J. Yang and G. Bai, "Finger-vein image restoration based on skin optical property," in *Proc. IEEE 11th Int. Conf. Signal Process.*, vol. 1, Oct. 2012, pp. 749–752.
- [43] J. Yang, B. Zhang, and Y. Shi, "Scattering removal for finger-vein image restoration," *Sensors*, vol. 12, no. 3, pp. 3627–3640, 2012.
- [44] J. Yang and Y. Shi, "Towards finger-vein image restoration and enhancement for finger-vein recognition," *Inf. Sci.*, vol. 268, pp. 33–52, Jun. 2014.
- [45] J. Yang, Y. Shi, and R. Wu, "Finger-vein recognition based on Gabor features," in *Biometric Systems, Design and Applications*, 2011, pp. 17–32.
- [46] J. Yang and Y. Shi, "Finger-vein ROI localization and vein ridge enhancement," *Pattern Recognit. Lett.*, vol. 33, no. 12, pp. 1569–1579, 2012.
- [47] J. Yang and Y. Shi, "Finger-vein network enhancement and segmentation," *Pattern Anal. Appl.*, vol. 17, no. 4, pp. 783–797, Nov. 2014.
- [48] W. You, W. Zhou, J. Huang, F. Yang, Y. Liu, and Z. Chen, "A bilayer image restoration for finger vein recognition," *Neurocomputing*, vol. 348, pp. 54–65, Jul. 2019.
- [49] T. Liu, J. Xie, W. Yan, P. Li, and H. Lu, "Finger-vein pattern restoration with direction-variance-boundary constraint search," *Eng. Appl. Artif. Intell.*, vol. 46, pp. 131–139, Nov. 2015.
- [50] J. Yang and X. Li, "Efficient finger vein localization and recognition," in *Proc. 20th Int. Conf. Pattern Recognit.*, Aug. 2010, pp. 1148–1151.
- [51] J. Yang, J. Wei, and Y. Shi, "Accurate ROI localization and hierarchical hyper-sphere model for finger-vein recognition," *Neurocomputing*, vol. 328, pp. 171–181, Feb. 2019.
- [52] L. Yang, G. Yang, Y. Yin, and R. Xiao, "Sliding window-based region of interest extraction for finger vein images," *Sensors*, vol. 13, no. 3, pp. 3799–3815, 2013.
- [53] S. Qiu *et al.*, "Finger-vein recognition based on dual-sliding window localization and pseudo-elliptical transformer," *Expert Syst. Appl.*, vol. 64, pp. 618–632, Dec. 2016.
- [54] M. Wang and D. Tang, "Region of interest extraction for finger vein images with less information losses," *Multimedia Tools Appl.*, vol. 76, no. 13, pp. 14937–14949, Jul. 2017.
- [55] H. Lu, Y. Wang, R. Gao, C. Zhao, and Y. Li, "A novel ROI extraction method based on the characteristics of the original finger vein image," *Sensors*, vol. 21, no. 13, p. 4402, Jun. 2021.
- [56] E. C. Lee, H. C. Lee, and K. R. Park, "Finger vein recognition using minutia-based alignment and local binary pattern-based feature extraction," *Int. J. Imag. Syst. Technol.*, vol. 19, no. 3, pp. 179–186, 2009.
- [57] Y. Lu, S. J. Xie, S. Yoon, J. Yang, and D. S. Park, "Robust finger vein ROI localization based on flexible segmentation," *Sensors*, vol. 13, no. 11, pp. 14339–14366, 2013.
- [58] J. Peng *et al.*, "An effective preprocessing method for finger vein recognition," *Proc. SPIE*, vol. 8878, Jul. 2013, Art. no. 887808.
- [59] K. Y. Shin, Y. H. Park, D. T. Nguyen, and R. P. Kang, "Finger-vein image enhancement using a fuzzy-based fusion method with Gabor and Retinex filtering," *Sensors*, vol. 14, no. 2, pp. 3095–3129, 2014.
- [60] Y. Liu *et al.*, "An improved finger-vein recognition algorithm based on template matching," *Proc. SPIE*, vol. 10157, Oct. 2016, Art. no. 101571T.
- [61] L. Yang, G. Yang, L. Zhou, and Y. Yin, "Superpixel based finger vein ROI extraction with sensor interoperability," in *Proc. Int. Conf. Biometrics (ICB)*, May 2015, pp. 444–451.
- [62] Q. Chen, L. Yang, G. Yang, and Y. Yin, "Geometric shape analysis based finger vein deformation detection and correction," *Neurocomputing*, vol. 311, pp. 112–125, Oct. 2018.
- [63] Z. Lu, S. Ding, and J. Yin, "Finger vein recognition based on finger crease location," *J. Electron. Imag.*, vol. 25, no. 4, Jul. 2016, Art. no. 043004.

- [64] S. Juan Xie, J. Yang, S. Yoon, L. Yu, and D. Sun Park, "Guided Gabor filter for finger vein pattern extraction," in *Proc. 8th Int. Conf. Signal Image Technol. Internet Based Syst.*, Nov. 2012, pp. 118–123.
- [65] P. Normakristagaluh, L. Spreeuwiers, and R. N. J. Veldhuis, "Finger-vein pattern recognition based on ICP on contours," in *Proc. WIC Symp. Inf. Theory*. Leuven, Belgium: Katholieke Universiteit Leuven, 2019, pp. 97–101.
- [66] J. Zhang, Z. Lu, and M. Li, "Finger-vein recognition based on gradient distribution and self-adaptive recovery model," *J. Electron. Imag.*, vol. 27, no. 5, 2018, Art. no. 053022.
- [67] W. Pi, J. Shin, and D. Park, "An effective quality improvement approach for low quality finger vein image," in *Proc. Int. Conf. Electron. Inf. Eng.*, Aug. 2010, pp. 424–427.
- [68] C. Liu, "A new finger vein feature extraction algorithm," in *Proc. 6th Int. Congr. Image Signal Process. (CISP)*, Dec. 2013, pp. 395–399.
- [69] N. Mahri, S. A. Sudi Suandi, and B. A. Rosdi, "Finger vein recognition algorithm using phase only correlation," in *Proc. Int. Workshop Emerg. Techn. Challenges Hand-Based Biometrics*, Aug. 2010, pp. 1–6.
- [70] Y. Lu, S. Yoon, and D. S. Park, "Finger vein recognition based on matching score-level fusion of Gabor features," *J. Korean Inst. Commun. Inf. Sci.*, vol. 38A, no. 2, pp. 174–182, Feb. 2013.
- [71] C. Kauba, E. Piciuccio, E. Maiorana, P. Campisi, and A. Uhl, "Advanced variants of feature level fusion for finger vein recognition," in *Proc. Int. Conf. Biometrics Special Interest Group (BIOSIG)*, Sep. 2016, pp. 1–7.
- [72] Y. Lu, S. J. Xie, S. Yoon, and D. S. Park, "Finger vein identification using polydirectional local line binary pattern," in *Proc. Int. Conf. ICT Converg. (ICTC)*, Oct. 2013, pp. 61–65.
- [73] C.-H. Hsia, J.-M. Guo, and C.-S. Wu, "Finger-vein recognition based on parametric-oriented corrections," *Multimedia Tools Appl.*, vol. 76, no. 23, pp. 25179–25196, Dec. 2017.
- [74] S. S. Jumaa and K. A. Zidan, "Finger vein recognition using two parallel enhancement approaches based fuzzy histogram equalization," *Periodicals Eng. Natural Sci.*, vol. 7, no. 1, pp. 514–529, 2019.
- [75] A. Banerjee, S. Basu, M. Nasipuri, and S. Basu, "ARTEM: A new system for human authentication using finger vein images," *Multimedia Tools Appl.*, vol. 77, no. 5, pp. 5857–5884, 2017.
- [76] E. C. Lee, H. Jung, and D. Kim, "New finger biometric method using near infrared imaging," *Sensors*, vol. 11, no. 3, pp. 2319–2333, 2011.
- [77] K. Wang, H. Ma, O. P. Popoola, and X. Li, "A novel finger vein pattern extraction method using oriented filtering technology," in *Proc. 8th World Congr. Intell. Control Autom.*, Jul. 2010, pp. 6240–6244.
- [78] L. Lei, F. Xi, and S. Chen, "Finger-vein image enhancement based on pulse coupled neural network," *IEEE Access*, vol. 7, pp. 57226–57237, 2019.
- [79] H. G. Kim *et al.*, "Illumination normalization for SIFT based finger vein authentication," in *Proc. Int. Symp. Vis. Comput.* Berlin, Germany: Springer, 2012, pp. 21–30.
- [80] M. Vlachos and E. Dermatas, "Finger vein segmentation from infrared images based on a modified separable Mumford Shah model and local entropy thresholding," *Comput. Math. Methods Med.*, vol. 2015, May 2015, Art. no. 868493.
- [81] S. Xie, Y. Lu, S. Yoon, J. Yang, and D. Park, "Intensity variation normalization for finger vein recognition using guided filter based single scale Retinex," *Sensors*, vol. 15, no. 7, pp. 17089–17105, 2015.
- [82] T. Liu, J. B. Xie, W. Yan, P. Q. Li, and H. Z. Lu, "An algorithm for finger-vein segmentation based on modified repeated line tracking," *Imag. Sci. J.*, vol. 61, no. 6, pp. 491–502, 2013.
- [83] H. Qin, L. Qin, and C. Yu, "Region growth-based feature extraction method for finger-vein recognition," *Opt. Eng.*, vol. 50, no. 5, 2011, Art. no. 057208.
- [84] N. Miura, A. Nagasaka, and T. Miyatake, "Extraction of finger-vein patterns using maximum curvature points in image profiles," *IEICE Trans. Inf. Syst.*, vol. 90, no. 8, pp. 1185–1194, 2007.
- [85] M. A. Syarif, T. S. Ong, A. B. J. Teoh, and C. Tee, "Enhanced maximum curvature descriptors for finger vein verification," *Multimedia Tools Appl.*, vol. 76, no. 5, pp. 6859–6887, 2016.
- [86] L. Yang, G. Yang, Y. Yin, and X. Xi, "Finger vein recognition with anatomy structure analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 8, pp. 1892–1905, Aug. 2018.
- [87] H. Qin, L. Qin, L. Xue, X. He, C. Yu, and X. Liang, "Finger-vein verification based on multi-features fusion," *Sensors*, vol. 13, no. 11, pp. 15048–15067, 2013.
- [88] J. Yang, Y. Shi, and G. Jia, "Finger-vein image matching based on adaptive curve transformation," *Pattern Recognit.*, vol. 66, pp. 34–43, Jun. 2017.
- [89] W. Song, T. Kim, H. Kim, J. Choi, H. Kong, and S. Lee, "A finger-vein verification system using mean curvature," *Pattern Recognit. Lett.*, vol. 32, no. 11, pp. 1541–1547, Aug. 2011.
- [90] H. Qin, X. He, X. Yao, and H. Li, "Finger-vein verification based on the curvature in Radon space," *Expert Syst. Appl.*, vol. 82, pp. 151–161, Oct. 2017.
- [91] Q. Yao, D. Song, X. Xu, and K. Zou, "A novel finger vein recognition method based on aggregation of radon-like features," *Sensors*, vol. 21, no. 5, p. 1885, Mar. 2021.
- [92] J. Zhang, Z. Lu, and M. Li, "Active contour-based method for finger-vein image segmentation," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 11, pp. 8656–8665, Nov. 2020.
- [93] C. B. Yu, H. F. Qin, and Y. Z. Cui, "Finger-vein image recognition combining modified Hausdorff distance with minutiae feature matching," *Interdiscipl. Sci. Comput. Life Sci.*, vol. 1, no. 4, pp. 280–289, 2009.
- [94] F. Liu, G. Yang, Y. Yin, and S. Wang, "Singular value decomposition based minutiae matching method for finger vein recognition," *Neurocomputing*, vol. 145, pp. 75–89, Dec. 2014.
- [95] L. Yang, G. Yang, X. Xi, X. Meng, C. Zhang, and Y. Yin, "Tri-branch vein structure assisted finger vein recognition," *IEEE Access*, vol. 5, pp. 21020–21028, 2017.
- [96] H. Khusnuliawati, C. Fatichah, and R. Soelaiman, "Multi-feature fusion using SIFT and LEBP for finger vein recognition," *Telkomnika*, vol. 15, no. 1, pp. 478–485, 2017.
- [97] T. D. Pham, Y. H. Park, D. T. Nguyen, S. Y. Kwon, and K. R. Park, "Nonintrusive finger-vein recognition system using NIR image sensor and accuracy analyses according to various factors," *Sensor*, vol. 15, no. 7, pp. 16866–16894, Jul. 2015.
- [98] H. C. Lee, B. J. Kang, E. C. Lee, and K. R. Park, "Finger vein recognition using weighted local binary pattern code based on a support vector machine," *J. Zhejiang Univ. Sci. C*, vol. 11, no. 7, pp. 514–524, Jul. 2010.
- [99] B. A. Rosdi, W. S. Chai, and S. A. Suandi, "Finger vein recognition using local line binary pattern," *Sensors*, vol. 11, no. 12, pp. 11357–11371, 2011.
- [100] A. William, T. S. Ong, S. H. Lau, and M. K. O. Goh, "Finger vein verification using local histogram of hybrid texture descriptors," in *Proc. IEEE Int. Conf. Signal Image Process. Appl. (ICSIPA)*, Oct. 2015, pp. 304–308.
- [101] T. S. Ong, A. William, T. Connie, and M. Kah Ong Goh, "Robust hybrid descriptors for multi-instance finger vein recognition," *Multimedia Tools Appl.*, vol. 77, no. 21, pp. 29163–29191, Nov. 2018.
- [102] G. Yang, X. Xi, and Y. Yin, "Finger vein recognition based on a personalized best bit map," *Sensors*, vol. 12, no. 2, pp. 1738–1757, 2012.
- [103] G. Yang *et al.*, "Finger vein recognition based on personalized weight maps," *Sensors*, vol. 8, no. 6, pp. 12093–12112, 2013.
- [104] X. Li, X. Xi, Y. Yin, and G. Yang, "Finger vein recognition based on personalized discriminative bit map," *Appl. Math. Inf. Sci.*, vol. 8, no. 6, pp. 3121–3127, Nov. 2014.
- [105] P. Sikarwar, "Finger vein recognition using local directional pattern," in *Proc. Int. Conf. Inventive Comput. Technol. (ICICT)*, vol. 1, Aug. 2016, pp. 1–5.
- [106] X. Meng, G. Yang, Y. Yin, and R. Xiao, "Finger vein recognition based on local directional code," *Sensors*, vol. 12, no. 11, pp. 14937–14952, 2012.
- [107] Y. Lu, S. Wu, Z. Fang, N. Xiong, S. Yoon, and D. S. Park, "Exploring finger vein based personal authentication for secure IoT," *Future Generat. Comput. Syst.*, vol. 77, pp. 149–160, Dec. 2017.
- [108] Y. Lu, S. Yoon, S. Wu, and D. S. Park, "Pyramid histogram of double competitive pattern for finger vein recognition," *IEEE Access*, vol. 6, pp. 56445–56456, 2018.
- [109] J. Yang, Y. Shi, and J. Yang, "Personal identification based on finger-vein features," *Comput. Hum. Behav.*, vol. 27, no. 5, pp. 1565–1570, 2011.
- [110] W. Yang, X. Huang, and Q. Liao, "Fusion of finger vein and finger dorsal texture for personal identification based on comparative competitive coding," in *Proc. 19th IEEE Int. Conf. Image Process.*, Sep. 2012, pp. 1141–1144.
- [111] J.-D. Wu and S.-H. Ye, "Driver identification using finger-vein patterns with Radon transform and neural network," *Expert Syst. Appl.*, vol. 36, no. 3, pp. 5793–5799, Apr. 2009.
- [112] D. Wang, J. Li, and G. Memik, "User identification based on finger-vein patterns for consumer electronics devices," *IEEE Trans. Consum. Electron.*, vol. 56, no. 2, pp. 799–804, May 2010.

- [113] R. P. Kang, "Finger vein recognition by combining global and local features based on SVM," *Comput. Inform.*, vol. 30, no. 2, pp. 295–309, Jan. 2012.
- [114] M. Mohan *et al.*, "Finger vein recognition using integrated responses of texture features," in *Proc. 4th Int. Work Conf. Bioinspired Intell. (IWobi)*, Jun. 2015, pp. 209–214.
- [115] F. Liu *et al.*, "Finger-vein recognition based on fusion of pixel level feature and super-pixel level feature," in *Proc. Chin. Conf. Biometric Recognit.*, vol. 8232. Cham, Switzerland: Springer, 2013, pp. 274–281.
- [116] F. Liu, Y. Yin, G. Yang, L. Dong, and X. Xi, "Finger vein recognition with superpixel-based features," in *Proc. IEEE Int. Joint Conf. Biometrics*, Sep. 2014, pp. 1–8.
- [117] L. Zhou, G. Yang, Y. Yin, L. Yang, and K. Wang, "Finger vein recognition based on stable and discriminative superpixels," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 30, no. 6, Jul. 2016, Art. no. 1650015.
- [118] L. Yang, G. Yang, Y. Yin, and X. Xi, "Exploring soft biometric trait with finger vein recognition," *Neurocomputing*, vol. 135, pp. 218–228, Jul. 2014.
- [119] F. Guan, K. Wang, and Q. Wu, "Bi-directional weighted modular B2DPCA for finger vein recognition," in *Proc. 3rd Int. Congr. Image Signal Process.*, vol. 1, Oct. 2010, pp. 93–97.
- [120] J.-D. Wu and C.-T. Liu, "Finger-vein pattern identification using SVM and neural network technique," *Expert Syst. Appl.*, vol. 38, no. 11, pp. 14284–14289, 2011.
- [121] F. Guan, K. Wang, and Q. Yang, "A study of two direction weighted (2D)²LDA for finger vein recognition," in *Proc. 4th Int. Congr. Image Signal Process.*, vol. 2, Oct. 2011, pp. 860–864.
- [122] S. Damavandinejadmonfared *et al.*, "Finger vein recognition using PCA-based methods," *World Acad. Sci., Eng. Technol.*, vol. 6, no. 6, pp. 1079–1081, 2012.
- [123] S. Damavandinejadmonfared, A. K. Mobarakeh, S. A. Suandi, and B. A. Rosdi, "Evaluate and determine the most appropriate method to identify finger vein," *Proc. Eng.*, vol. 41, pp. 516–521, Jan. 2012.
- [124] H. Liu, G. Yang, L. Yang, K. Su, and Y. Yin, "Anchor-based manifold binary pattern for finger vein recognition," *Sci. China Inf. Sci.*, vol. 62, no. 5, pp. 1–16, May 2019.
- [125] X. Xi, G. Yang, Y. Yin, and X. Meng, "Finger vein recognition with personalized feature selection," *Sensors*, vol. 13, no. 9, pp. 11243–11259, Sep. 2013.
- [126] X. Xi, L. Yang, and Y. Yin, "Learning discriminative binary codes for finger vein recognition," *Pattern Recognit.*, vol. 66, pp. 26–33, Jun. 2017.
- [127] H. Liu, L. Yang, G. Yang, and Y. Yin, "Discriminative binary descriptor for finger vein recognition," *IEEE Access*, vol. 6, pp. 5795–5804, 2018.
- [128] H. Liu, G. Yang, L. Yang, and Y. Yin, "Learning personalized binary codes for finger vein recognition," *Neurocomputing*, vol. 365, pp. 62–70, Nov. 2019.
- [129] K. Su, G. Yang, L. Yang, D. Li, P. Su, and Y. Yin, "Learning binary hash codes for finger vein image retrieval," *Pattern Recognit. Lett.*, vol. 117, pp. 74–82, Jan. 2019.
- [130] L. Yang, G. Yang, X. Xi, K. Su, Q. Chen, and Y. Yin, "Finger vein code: From indexing to matching," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 5, pp. 1210–1223, May 2018.
- [131] J. Zhang and P. Fang, "Two encryption schemes of finger vein template," in *Proc. 12th IEEE Int. Conf. Anti-Counterfeiting, Secur., Identificat. (ASID)*, Nov. 2018, pp. 169–173.
- [132] V. Krivokuca and S. Marcel, "On the recognition performance of biohash-protected finger vein templates," in *Handbook of Vascular Biometrics* (Advances in Computer Vision and Pattern Recognition). 2020, pp. 465–467.
- [133] L. Shao, H. Ren, L. Sun, C. Han, and J. Guo, "Template protection based on chaotic map for finger vein recognition," *IEEE Trans. Electr. Electron. Eng.*, vol. 17, no. 1, pp. 82–95, Jan. 2022.
- [134] Y. Liu, J. Ling, Z. Liu, J. Shen, and C. Gao, "Finger vein secure biometric template generation based on deep learning," *Soft Comput.*, vol. 22, no. 7, pp. 2257–2265, Apr. 2018.
- [135] W. Yang, Q. Rao, and Q. Liao, "Personal identification for single sample using finger vein location and direction coding," in *Proc. Int. Conf. Hand-Based Biometrics*, Nov. 2011, pp. 176–181.
- [136] P. Gupta and P. Gupta, "An accurate finger vein based verification system," *Digit. Signal Process.*, vol. 38, pp. 43–52, Mar. 2015.
- [137] J. Yang and Y. Jia, "A method of multispectral finger-vein image fusion," in *Proc. IEEE 11th Int. Conf. Signal Process.*, vol. 1, Oct. 2012, pp. 753–756.
- [138] J. Yang, M. Wu, W. Wang, and Y. Shi, "Finger-vein matching based on adaptive vector field estimation," in *Proc. 19th IEEE Int. Conf. Image Process.*, Sep. 2012, pp. 145–148.
- [139] T. S. Beng and B. A. Rosdi, "Finger-vein identification using pattern map and principal component analysis," in *Proc. IEEE Int. Conf. Signal Image Process. Appl. (ICSIPA)*, Nov. 2011, pp. 530–534.
- [140] J. Peng, Q. Li, A. A. Abd El-Latif, N. Wang, and X. Niu, "Finger vein recognition with Gabor wavelets and local binary patterns," *IEICE Trans. Inf. Syst.*, vol. 96, no. 8, pp. 1886–1889, Aug. 2013.
- [141] Y. Yang, G. Yang, and S. Wang, "Finger vein recognition based on multi-instance," *Int. J. Digit. Content Technol. Appl.*, vol. 6, no. 11, pp. 86–94, Jun. 2012.
- [142] C.-C. Chiu, T.-K. Liu, W.-T. Lu, W.-P. Chen, and J.-H. Chou, "A micro-control capture images technology for the finger vein recognition based on adaptive image segmentation," *Microsyst. Technol.*, vol. 24, no. 10, pp. 4165–4178, Oct. 2018.
- [143] W. Kang, Y. Lu, D. Li, and W. Jia, "From noise to feature: Exploiting intensity distribution as a novel soft biometric trait for finger vein recognition," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 4, pp. 858–869, Apr. 2018.
- [144] S. J. Xie, S. Yoon, J. Yang, Y. Lu, D. S. Park, and B. Zhou, "Feature component-based extreme learning machines for finger vein recognition," *Cognit. Comput.*, vol. 6, no. 3, pp. 446–461, Sep. 2014.
- [145] G. Yang, X. Xi, and Y. Yin, "Finger vein recognition based on (2D)² PCA and metric learning," *BioMed Res. Int.*, vol. 2012, May 2012, Art. no. 324249.
- [146] D. Cao, J. Yang, Y. Shi, and C. Xu, "Structure feature extraction for finger-vein recognition," in *Proc. 2nd IAPR Asian Conf. Pattern Recognit.*, Nov. 2013, pp. 567–571.
- [147] H. Qin and M. A. El-Yacoubi, "Finger-vein quality assessment by representation learning from binary images," in *Proc. Int. Conf. Neural Inf. Process.*, vol. 9489. Cham, Switzerland: Springer, 2015, pp. 421–431.
- [148] H. Qin and M. A. El-Yacoubi, "Deep representation for finger-vein image-quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 8, pp. 1677–1693, Aug. 2018.
- [149] J. Zeng, Y. Chen, and C. Qin, "Finger-vein image quality assessment based on light-CNN," in *Proc. 14th IEEE Int. Conf. Signal Process. (ICSP)*, Aug. 2018, pp. 768–773.
- [150] X.-J. Guo, D. Li, H.-G. Zhang, and J.-F. Yang, "Image restoration of finger-vein networks based on encoder-decoder model," *Optoelectron. Lett.*, vol. 15, no. 6, pp. 463–467, Nov. 2019.
- [151] J. Choi, K. J. Noh, S. W. Cho, S. H. Nam, M. Owais, and K. R. Park, "Modified conditional generative adversarial network-based optical blur restoration for finger-vein recognition," *IEEE Access*, vol. 8, pp. 16281–16301, 2020.
- [152] H. Qin and M. A. El-Yacoubi, "Deep representation-based feature extraction and recovering for finger-vein verification," *IEEE Trans. Inf. Forensics Security*, vol. 12, no. 8, pp. 1816–1829, Aug. 2017.
- [153] H. Qin and P. Wang, "Finger-vein verification based on LSTM recurrent neural networks," *Appl. Sci.*, vol. 9, no. 8, p. 1687, 2019.
- [154] B. Hou and R. Yan, "Convolutional auto-encoder based deep feature learning for finger-vein verification," in *Proc. IEEE Int. Symp. Med. Meas. Appl. (MeMeA)*, Jun. 2018, pp. 1–5.
- [155] B. Hou and R. Yan, "Convolutional autoencoder model for finger-vein verification," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 5, pp. 2067–2074, May 2020.
- [156] E. Jalilian and A. Uhl, "Finger-vein recognition using deep fully convolutional neural semantic segmentation networks: The impact of training data," in *Proc. IEEE Int. Workshop Inf. Forensics Secur. (WIFS)*, Dec. 2018, pp. 1–8.
- [157] W. Yang, C. Hui, Z. Chen, J.-H. Xue, and Q. Liao, "FV-GAN: Finger vein representation using generative adversarial networks," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 9, pp. 2512–2524, Sep. 2019.
- [158] J. M. Song, W. Kim, and K. R. Park, "Finger-vein recognition based on deep DenseNet using composite image," *IEEE Access*, vol. 7, pp. 66845–66863, 2019.
- [159] J. Zeng *et al.*, "Finger vein verification algorithm based on fully convolutional neural network and conditional random field," *IEEE Access*, vol. 8, pp. 65402–65419, 2020.
- [160] S. A. Radzi, M. K. Hani, and R. Bakhteri, "Finger-vein biometric identification using convolutional neural network," *Turkish J. Electr. Eng. Comput. Sci.*, vol. 24, no. 3, pp. 1863–1878, 2016.
- [161] M. Fayyaz, M. Hajizadeh-Saffar, M. Sabokrou, M. Hoseini, and M. Fathy, "A novel approach for finger vein verification based on self-taught learning," in *Proc. 9th Iranian Conf. Mach. Vis. Image Process. (MVIP)*, Nov. 2015, pp. 88–91.

- [162] G. Meng, P. Fang, and B. Zhang, "Finger vein recognition based on convolutional neural network," in *Proc. MATEC Web Conf.*, vol. 128, 2017, Art. no. 04015.
- [163] R. Das, E. Piciuccio, E. Maiorana, and P. Campisi, "Convolutional neural network for finger-vein-based biometric identification," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 2, pp. 360–373, Feb. 2019.
- [164] Z. Tao, H. Wang, Y. Hu, Y. Han, S. Lin, and Y. Liu, "DGLFV: Deep generalized label algorithm for finger-vein recognition," *IEEE Access*, vol. 9, pp. 78594–78606, 2021.
- [165] H. Hong, M. Lee, and K. Park, "Convolutional neural network-based finger-vein recognition using NIR image sensors," *Sensors*, vol. 17, no. 6, p. 1297, Jun. 2017.
- [166] Y. Lu, S. Xie, and S. Wu, "Exploring competitive features using deep convolutional neural network for finger vein recognition," *IEEE Access*, vol. 7, pp. 35113–35123, 2019.
- [167] S. Li, H. Zhang, and J. Yang, "Finger vein recognition based on local graph structural coding and CNN," *Proc. SPIE*, vol. 11069, May 2019, Art. no. 1106931.
- [168] G. Wang, C. Sun, and A. Sowmya, "Multi-weighted co-occurrence descriptor encoding for vein recognition," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 375–390, 2020.
- [169] J. Zhang, Z. Lu, M. Li, and H. Wu, "GAN-based image augmentation for finger-vein biometric recognition," *IEEE Access*, vol. 7, pp. 183118–183132, 2019.
- [170] H. Huang, S. Liu, H. Zheng, L. Ni, Y. Zhang, and W. Li, "DeepVein: Novel finger vein verification methods based on deep convolutional neural networks," in *Proc. IEEE Int. Conf. Identity, Secur. Behav. Anal. (ISBA)*, Feb. 2017, pp. 1–8.
- [171] S. Tang, S. Zhou, W. Kang, Q. Wu, and F. Deng, "Finger vein verification using a Siamese CNN," *IET Biometrics*, vol. 8, no. 5, pp. 306–315, Sep. 2019.
- [172] Y. Fang, Q. Wu, and W. Kang, "A novel finger vein verification system based on two-stream convolutional network learning," *Neurocomputing*, vol. 290, pp. 100–107, May 2018.
- [173] H. Hu *et al.*, "FV-Net: Learning a finger-vein feature representation based on a CNN," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 3489–3494.
- [174] D. Zhao, H. Ma, Z. Yang, J. Li, and W. Tian, "Finger vein recognition based on lightweight CNN combining center loss and dynamic regularization," *Infr. Phys. Technol.*, vol. 105, Mar. 2020, Art. no. 103221.
- [175] B. Hou and R. Yan, "ArcVein-arccosine center loss for finger vein verification," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.
- [176] W.-F. Ou, L.-M. Po, C. Zhou, Y. A. U. Rehman, P.-F. Xian, and Y.-J. Zhang, "Fusion loss and inter-class data augmentation for deep finger vein feature learning," *Expert Syst. Appl.*, vol. 171, Jun. 2021, Art. no. 114584.
- [177] N. M. Kamaruddin and B. A. Rosdi, "A new filter generation method in PCANet for finger vein recognition," *IEEE Access*, vol. 7, pp. 132966–132978, 2019.
- [178] Y. Zhang, W. Li, L. Zhang, X. Ning, L. Sun, and Y. Lu, "Adaptive learning Gabor filter for finger-vein recognition," *IEEE Access*, vol. 7, pp. 159821–159830, 2019.
- [179] R. Li, Z. G. Su, H. G. Zhang, and J. F. Yang, "Application of improved GCNs in feature representation of finger-vein," *J. Signal Process.*, vol. 36, no. 4, pp. 550–561, 2020.
- [180] K. Wan, J. M. Song, and R. P. Kang, "Multimodal biometric recognition based on convolutional neural network by the fusion of finger-vein and finger shape using near-infrared (NIR) camera sensor," *Sensors*, vol. 18, no. 7, p. 2296, 2018.
- [181] N. Alay and H. H. Al-Baity, "Deep learning approach for multimodal biometric recognition system based on fusion of iris, face, and finger vein traits," *Sensors*, vol. 20, 2020.
- [182] B. A. El-Rahiem, F. E. A. El-Samie, and M. Amin, "Multimodal biometric authentication based on deep fusion of electrocardiogram (ECG) and finger vein," *Multimedia Syst.*, vol. 3, 2021.
- [183] C. Zhou *et al.*, "A hybrid fusion model of iris, palm vein and finger vein for multi-biometric recognition system," *Multimedia Tools Appl.*, vol. 79, no. 12, 2020.
- [184] T. Bunda, "3D point cloud reconstruction based on the finger vascular pattern," Univ. Twente, 2018.
- [185] W. Kang *et al.*, "Study of a full-view 3D finger vein verification technique," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 1175–1189, 2020.
- [186] Y. Zhan *et al.*, "3D finger vein biometric authentication with photoacoustic tomography," *Appl. Opt.*, vol. 59, no. 28, pp. 8751–8758, 2020.



Borui Hou received the B.S. degree from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2015, and the Ph.D. degree from Southeast University, Dhaka, Bangladesh, in 2021.

He joined the Shanghai Aerospace Electronics Technology Institute in 2021. His academic interests include fault diagnosis, signal analysis including physiological signal analysis, pattern identification, and computer vision.



Huijie Zhang is currently pursuing the Ph.D. degree with Southeast University, Nanjing, China.

His current research interests include deep learning, computer vision, image processing, biometric security, and pattern recognition.



Ruqiang Yan (Fellow, IEEE) received the M.S. degree in precision instrument and machinery from the University of Science and Technology of China, Hefei, China, in 2002, and the Ph.D. degree in mechanical engineering from the University of Massachusetts at Amherst, Amherst, MA, USA, in 2007.

He was a Guest Researcher with the National Institute of Standards and Technology (NIST) from 2006 to 2008 and a Professor with the School of Instrument Science and Engineering, Southeast University, Nanjing, China, from 2009 to 2018. He joined the School of Mechanical Engineering, Xi'an Jiaotong University, Xi'an, China, in 2018. His current research interests include data analytics, AI, and energy-efficient sensing and sensor networks for the condition monitoring and health diagnosis of large-scale, complex, dynamical systems.

Dr. Yan was a Fellow of ASME in 2019. His honors and awards include the IEEE Instrumentation and Measurement Society Technical Award in 2019, the New Century Excellent Talents in University Award from the Ministry of Education in China in 2009, and multiple best paper awards. He has been serving as an Editor-in-Chief for the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT since January 2022. He is an Associate Editor of the IEEE SENSORS JOURNAL and an Editorial Board Member of the *Chinese Journal of Mechanical Engineering*.